

NATIVE GEOMETRY DIFFUSION: DIFFUSABILITY-GUIDED COMPACT GEOMETRY LA- TENTS FOR JOINT RGB AND GEOMETRY GENERATION

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Paper under double-blind review

ABSTRACT

The latent space is a central design choice in modern visual diffusion, yet most latent spaces are optimized for single-image appearance rather than multi-view geometry, which limits their use for 3D-consistent generation and, in turn, as visual world models. Pixel VAEs are compact and decodable but weakly geometry-aware, whereas raw geometry-foundation features carry depth, pose, and correspondence but are high-dimensional, ill-conditioned, and spectrally near-degenerate as a diffusion state. We present *Native Geometry Diffusion* (NGD), a compact geometry representation autoencoder and flow-matching model for geometry-aware generation. NGD compresses frozen geometry-foundation features into a small latent whose generated states decode jointly into RGB and geometry, and we use spectral, semantic, and spatial diagnostics to show that this compact latent preserves useful geometry-native structure while greatly improving conditioning and semantic/spatial organization — separating reconstruction quality from latent diffusability. To make the latent usable as a controllable world-model state, we adapt flow matching with RAEv2-style x -prediction, train/inference-aligned reference conditioning that separates observed evidence from generated state, and metric ray-space pose conditioning that treats camera motion as a scale-aware control input. We evaluate the latent against pixel-VAE, semantic-RAE, and raw geometry-feature baselines with reconstruction, latent-space, and geometry-aware consistency metrics.

1 INTRODUCTION

Diffusion and flow-matching models have become the dominant paradigm for visual generation, producing high-fidelity images and increasingly realistic videos from text, images, and camera trajectories. They are also increasingly used as visual world models — generating what a scene looks like as an observer moves through it — which demands more than per-frame photorealism: the generated video must stay *3D-consistent* as the camera moves. Standard generators, built on pixel-oriented VAE latents with camera signals added from the outside, achieve this only weakly. We argue that 3D consistency is decided largely *before* the generator, by the latent space in which generation happens, with the camera-injection scheme playing only a supporting role; the latent is therefore our central concern. This view extends a recurring lesson — that moving from pixels to VAE latents, and then to semantically structured representations, repeatedly improved quality and training efficiency — from single-image synthesis, where it was established, to a setting where consistency must hold *across* views rather than within a single frame.

Improving the latent for geometry-aware generation requires moving beyond the single-image pixel VAE along *two independent axes*. The *view* axis is single-image versus multi-view encoding: a conventional VAE encodes each frame independently, so cross-view correspondence must be rediscovered by the generator, whereas a jointly encoded multi-view latent carries that correspondence in the representation itself. The *representation* axis is appearance versus geometry: a pixel VAE is optimized for per-pixel appearance and discards the rest, whereas a geometry-aware latent encodes depth, camera, and scene layout and can be decoded to geometry as well as RGB. The two

axes are complementary and neither suffices alone — a multi-view pixel VAE is still geometrically ungrounded, and a single-image geometry latent still cannot enforce cross-view consistency — so geometry-aware generation calls for a latent that is at once *multi-view* and *geometry-native*.

Most existing systems instead keep the appearance-centric, view-independent VAE latent and compensate from the outside: multi-view and video diffusion inject camera poses, depth maps, point clouds, or ray embeddings as conditions; geometry-aware video models add auxiliary prediction heads, correspondence losses, or post-training rewards; and 3D pipelines synthesize views and reconstruct a NeRF or 3D Gaussian representation afterward. These strategies help, but they share one limitation: geometry is imposed *around* a latent that was never designed to represent it, so 3D consistency must be enforced as an external constraint on an appearance-centric state — which is exactly what moving along both axes at once would avoid.

Geometry foundation models fit both axes naturally. DUST3R, VGGT, and Depth Anything 3 recover depth, camera motion, rays, and point clouds from sparse or monocular views, and their internal features are both multi-view (produced by cross-view attention) and geometry-native (trained for 3D reconstruction). Concurrent work realizes exactly this idea, diffusing directly in these raw features for pose-conditioned novel-view synthesis and reporting clear gains over pixel-VAE and semantic-RAE latents (Jang et al., 2026). This validates the destination but not the vehicle: the *raw features* are a poor generation space. They are high-dimensional and severely ill-conditioned, and although shallow layers retain strong cross-view correspondence, their channel spectrum is nearly degenerate and their per-view semantic and spatial organization is uneven. As a result, their seemingly low transport cost reflects spectral degeneracy rather than a genuinely smooth latent, and must be read together with rank and conditioning diagnostics (Section 4.1). Diffusing raw features also keeps the generator in a very large state, yet naively compressing them risks destroying the correspondence structure that made them attractive. The open question is therefore not *whether* geometry features can be a generation space, but how to obtain a *compact* one that stays multi-view, geometry-native, well-conditioned, structured, decodable, and usable as a controllable visual world-model state.

We answer it with a simple hypothesis: geometry-aware generation should happen in a multi-view, geometry-native latent, but one *explicitly designed to be diffusable* rather than used raw. We build a *compact* geometry representation autoencoder on a frozen geometry foundation model and train flow matching in its small latent. The frozen encoder supplies both axes, while the learned compression repairs the raw features — lowering dimensionality, conditioning the latent, and, guided by diffusability and structure diagnostics rather than reconstruction alone, restoring semantic and spatial organization. As Section 4.1 shows, no pixel-VAE, semantic-RAE, or raw geometry latent is good on all these axes at once, whereas the compact NGD latent is, and it stays jointly decodable to RGB and geometry.

We then adapt flow matching to this latent through three choices. First, following the RAEv2 observation that representation latents benefit from directly predicting the clean latent (Singh et al., 2026), we use an x -prediction parameterization, so the model predicts the same compact geometry state that the RGB/geometry decoder consumes. Second, we align reference conditioning between training and inference. It is common to place clean reference latents into the same flow state as the noisy target latents, which is fine when each image has a fixed per-frame code, but subtly wrong for a DA3-style geometry encoder: because cross-view attention makes each view’s feature depend on the others, the same reference frame is encoded differently alone than jointly with its neighbors.

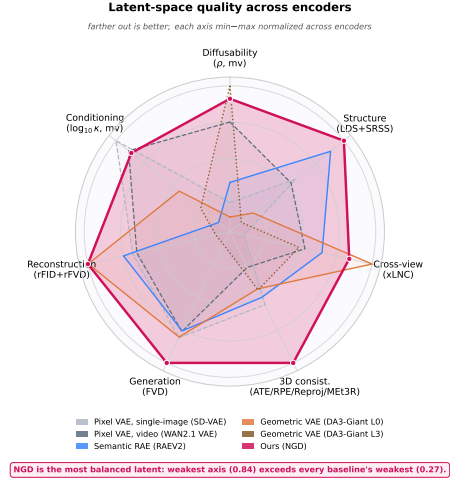


Figure 1: **Latent-space quality radar.**

Seven axes, min–max normalized across six encoders and oriented so farther out is better: diffusability (ρ), structure (LDS+SRSS), cross-view ($\times$ LNC*), 3D consistency (five VGGT metrics), generation (FVD), reconstruction (rFID+rFVD), and conditioning ($\log_{10} \kappa$). Every baseline collapses on some axis; NGD stays large and balanced.



Figure 2: **Overview.** *Left:* the two latent axes — single-image vs. multi-view, and appearance (pixel VAE) vs. geometry-native — placing pixel VAE, semantic RAE, raw geometry features, and NGD on the plane. *Right:* NGD encodes images through a frozen geometry foundation model and a compact geometry autoencoder, performs flow matching in the compact geometry latent with text, camera, and reference conditioning, and decodes generated latents into RGB and geometry.

Training could use the full ground-truth view set to obtain such a joint reference feature, but at inference the target views do not yet exist, so only a single-view encoding is available — a train/inference mismatch. NGD avoids this by separating observed evidence from generated state: all output slots, including reference-view slots, stay noisy flow states that are generated jointly, while the observed reference image enters only as a clean conditioning token. Third, since camera motion is the control input of a visual world model, we express it in metric ray space: building on ray-space camera conditioning (Yin et al., 2026), we inject metric pose through per-token rays rather than a global or per-scene-normalized pose code, preserving absolute translation scale across datasets. Together these choices let one backbone operate under text-only, camera-conditioned, and reference-conditioned regimes over a single flow state whose generated latents remain jointly decodable to RGB and geometry.

Our contributions are:

- **A compact, diffusable geometry latent (our central contribution).** We build a compact autoencoder over frozen geometry-foundation features and, using spectral, semantic, and spatial diagnostics, characterize what makes such a latent a good *generation* space. We find that raw geometry features preserve useful cross-view correspondence but remain high-dimensional, ill-conditioned, and spectrally near-degenerate; our autoencoder fuses DA3 levels 0–3 so that shallow correspondence and deeper geometry/semantic cues are distilled into one small, decodable, better-conditioned state. Unlike raw multi-level geometry diffusion, this latent is generated in one pass and decodes jointly into RGB and geometry.
- **World-model-oriented flow design for geometry latents.** On this latent we adapt flow matching through three choices — RAEv2-style x -prediction (Singh et al., 2026), train/inference-aligned reference conditioning that separates observed evidence from generated state, and metric ray-space pose conditioning — so a single backbone supports text-, camera-, and reference-conditioned generation as a controllable world-model state.

2 RELATED WORK

Latent spaces and latent diffusability. Latent diffusion made high-resolution generation practical by moving denoising from pixels to compressed VAE latents. More recent representation autoencoders (RAEs) replace the image VAE encoder with pretrained visual features, showing that the latent space is an active modeling choice rather than a neutral compression detail. RAEv2 studies practical RAE recipes, including multi-layer representations, REPA-style alignment, x -prediction, and internal guidance for improving the reconstruction-generation trade-off (Singh et al., 2026). Orthogonal to architecture, Diffusing in the Right Space argues that tokenizer reconstruction alone does not predict generation quality, and introduces diagnostics for latent diffusability, semantic separability, spatial structure, and spectral behavior (Zhong et al., 2026). NGD adopts this view, but moves

it from semantic image features to geometry latents: the latent must be not only reconstructive and diffusable, but also multi-view, geometry-decodable, and camera-controllable.

Geometry foundation models as generative states. Feed-forward geometry foundation models such as DUST3R, VGGT, Depth Anything 3, and related systems infer depth, camera motion, point maps, and correspondences from sparse or unposed views. Several recent works repurpose their internal representations as world-model states. VGGT-World predicts the temporal evolution of frozen VGGT features, and FlowWM performs stochastic flow matching in pretrained feature spaces for perception-oriented world modeling (Sun et al., 2026; Porcher et al., 2026). These works support the premise that pretrained geometry features can be more appropriate states than RGB pixels for scene dynamics. NGD differs in objective and interface: we build a compact generative autoencoder over geometry features, train diffusion in that compact latent, and decode generated states jointly to RGB and geometry for controllable visual synthesis.

Geometry-aware novel-view and scene generation. Novel-view synthesis and 3D scene generation require images that are consistent with a scene under camera motion. Classical methods reconstruct an explicit per-scene representation, whereas recent diffusion systems use image priors and inject camera, depth, point, or ray information as conditions. Gen3R connects feed-forward reconstruction models with video diffusion by aligning geometry latents from VGGT with appearance latents from a video generator, enabling joint RGB and geometry generation (Huang et al., 2026). Most directly related to us, concurrent work diffuses the raw multi-level features of a frozen geometry backbone for multi-view generation (Jang et al., 2026). We share its thesis that geometry representations are better generation spaces than pixel VAE latents, but not its choice of state: raw features are high-dimensional, strongly anisotropic, and ill-conditioned. NGD therefore learns one compact latent over the multi-level stack, trains a single diffusion model over that latent, and uses latent diagnostics rather than layer heuristics to select the representation.

Geometry consistency in video generation. Large video diffusion models can produce realistic motion while drifting in camera geometry or cross-view consistency. Existing approaches improve geometry either by adding external controls and auxiliary heads to an appearance-space generator, or by post-training with geometry-based rewards and consistency metrics. VGGTPO, for example, connects diffusion latents to geometry foundation models and optimizes camera-motion and reprojection rewards in latent space (An et al., 2026). NGD is complementary: instead of treating geometry as a reward or auxiliary signal around an appearance latent, we make the generated latent itself geometry-native from the start. This lets RGB quality, geometry decodability, and camera consistency be evaluated in one representation space.

Camera and reference conditioning. Camera-conditioned video and multi-view diffusion must specify where each view is. Ray-space methods such as RayPE encode per-pixel or per-token rays, often in Plücker form, so attention is addressed in camera-ray coordinates (Yin et al., 2026). We use ray-space conditioning, but train it with a geometry-native latent and preserve metric translation scale when metric camera sidecars are available. Reference conditioning raises a separate issue. In image-to-video and novel-view diffusion, known views are often mixed with noisy target states; for set-based geometry encoders this creates a mismatch because a reference feature encoded with all target views during training cannot be computed at inference. NGD separates the two roles: all output slots are noisy flow states, while observed references are clean conditioning tokens. This formulation is related in spirit to conditional diffusion and inpainting, but is specialized to geometry-latent multi-view generation where the mismatch is visible in depth, ray, reprojection, and point-cloud consistency.

3 METHOD

Native Geometry Diffusion (NGD) makes the latent space, rather than an auxiliary conditioning signal, carry the 3D inductive bias. Given a set of views $\mathbf{I}_{1:V}$, optional cameras $(\mathbf{K}_{1:V}, \mathbf{T}_{1:V})$, text condition c , and optional reference views, NGD follows the pipeline

$$\mathbf{I}_{1:V} \xrightarrow{\mathcal{E}} \mathbf{F}_{1:V}^{0:3} \xrightarrow{\mathcal{A}_\phi} \mathbf{z}_{1:V} \xrightarrow{\text{flow}} \hat{\mathbf{z}}_{1:V} \xrightarrow{\mathcal{D}_\theta} (\hat{\mathbf{I}}_{1:V}, \hat{\mathbf{G}}_{1:V}). \quad (1)$$

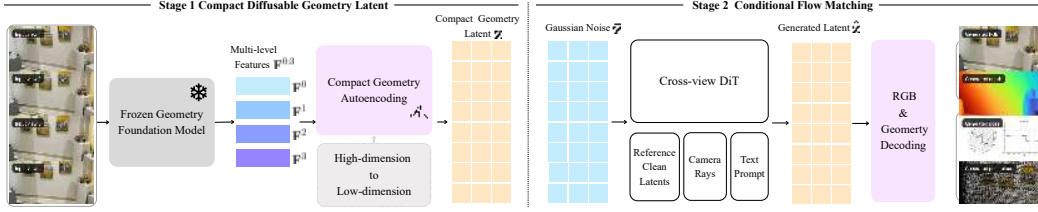


Figure 3: **Overview of Native Geometry Diffusion (NGD).** Stage 1 distills frozen multi-level geometry foundation features into a compact, diffusible geometry latent. Stage 2 performs conditional flow matching only over generated-view latents, while clean reference latents, camera rays, and text act as controls. The final latent is decoded jointly into RGB, depth, camera pose, and point maps.

Here $\mathcal{E}$ is a frozen geometry foundation encoder, $\mathcal{A}_\phi$ is a compact geometry autoencoder, and $\mathcal{D}_\theta$ decodes generated latents into RGB and geometry. The key distinction from pixel-latent diffusion is that $\mathbf{z}$ is not an RGB compression code: it is distilled from multi-view, geometry-native foundation features and remains decodable to depth, rays, and derived point maps. The key distinction from raw-feature geometry diffusion is that the generator never operates in the high-dimensional, ill-conditioned foundation feature space; it sees a compact and standardized state.

We use three symbols throughout. $\mathbf{z}$ denotes the codec latent, $\bar{\mathbf{z}}$ its standardized version used by flow matching, and $\bar{\mathbf{z}}^{\text{ref}}$ the clean reference evidence encoded from the observed views only. NGD keeps these roles separate: generated views are ODE states, references are conditioning evidence, and text/cameras are controls.

3.1 A COMPACT DIFFUSABLE GEOMETRY LATENT

The central representation problem is to retain the correspondence and geometry encoded by a foundation model while removing the properties that make raw features poor diffusion states: excessive channel count, anisotropy, and poor conditioning. NGD therefore learns a compact autoencoder over a frozen DA3 feature hierarchy rather than selecting a single raw layer to diffuse.

Multi-level geometry features. For each view v , the frozen DA3 encoder $\mathcal{E}$ produces patch tokens $\mathbf{f}_{v,\ell} \in \mathbb{R}^{N \times C_\ell}$ at four internal levels $\ell \in \{0, 1, 2, 3\}$, where $N = h_p w_p$ is the patch grid size. We remove class tokens, reshape patch tokens into feature maps, and concatenate the hierarchy:

$$\mathbf{X}_v = \parallel_{\ell=0}^3 \text{reshape}(\mathbf{f}_{v,\ell}) \in \mathbb{R}^{C_x \times h_p \times w_p}, \quad C_x = \sum_{\ell=0}^3 C_\ell. \quad (2)$$

Level 0 carries local appearance and correspondence; deeper levels carry more global geometry and semantic information. We normalize each level with precomputed per-channel statistics before fusion,

$$\tilde{\mathbf{X}}_v = \text{concat}_\ell \left(\frac{\mathbf{X}_{v,\ell} - \boldsymbol{\mu}_\ell}{\boldsymbol{\sigma}_\ell + \epsilon} \right). \quad (3)$$

Autoencoding. The codec $\mathcal{A}_\phi$ maps normalized multi-level features to a grid-shaped latent $\mathbf{z}_v \in \mathbb{R}^{C_z \times h \times w}$:

$$(\boldsymbol{\mu}_v, \log \boldsymbol{\sigma}_v^2) = \text{Enc}_\phi(\tilde{\mathbf{X}}_v), \quad (4)$$

$$\mathbf{z}_v = \boldsymbol{\mu}_v + \boldsymbol{\epsilon}_v \odot \exp(\tfrac{1}{2} \log \boldsymbol{\sigma}_v^2), \quad \boldsymbol{\epsilon}_v \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (5)$$

$$\hat{\mathbf{X}}_v = \text{Dec}_\phi(\mathbf{z}_v). \quad (6)$$

In the main models $C_z = 128$ and $h = w = 18$ at 252×252 input resolution. At diffusion training and inference time we use the posterior mean $\boldsymbol{\mu}_v$; the KL term is kept small so the bottleneck behaves as a stable representation autoencoder rather than a heavily regularized generative VAE. Architecturally, the encoder and decoder are lightweight convolutional pyramids with spatial self-attention blocks, preserving the DA3 patch layout while allowing long-range spatial mixing.

Joint RGB and geometry decoding. The decoder output is split and denormalized back into $\{\hat{\mathbf{f}}_{v,\ell}\}_{\ell=0}^3$. An RGB head $\mathcal{D}_\theta^{\text{rgb}}$ maps decoder tokens to images, while the reconstructed feature hierarchy is passed through the frozen DA3 dense prediction head to obtain geometry:

$$\hat{\mathbf{I}}_v = \mathcal{D}_\theta^{\text{rgb}}(\mathbf{z}_v), \quad \hat{\mathbf{G}}_v = \mathcal{H}_{\text{DPT}}(\{\hat{\mathbf{f}}_{v,\ell}\}_{\ell=0}^3). \quad (7)$$

Thus RGB and geometry originate from the same compact state. For raw single-layer DA3 base-lines, where only one foundation-model layer is synthesized, we recover deeper features with frozen backbone propagation before DPT decoding; NGD itself decodes the full multi-level stack from its compact bottleneck.

Diffusability-guided codec objective. The codec is not selected by reconstruction quality alone. We train it with

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_{\text{feat}} + \lambda_{\text{kl}}\mathcal{L}_{\text{kl}} + \mathcal{L}_{\text{rgb}} + \mathcal{L}_{\text{geo}} + \mathcal{L}_{\text{repr}}, \quad (8)$$

where $\mathcal{L}_{\text{feat}}$ is a weighted level-wise feature reconstruction loss, $\mathcal{L}_{\text{rgb}}$ contains RGB reconstruction and perceptual terms, $\mathcal{L}_{\text{geo}}$ supervises DPT depth/ray outputs against pseudo-ground-truth outputs from real frozen features, and $\mathcal{L}_{\text{repr}}$ contains REPA-style token alignment and relational structure distillation (Singh et al., 2026; Zhong et al., 2026). The last term is important: it discourages compressions that minimize feature error while destroying semantic and spatial neighborhoods, a failure mode we observe in raw geometry features. After training, we freeze the codec and standardize latents channel-wise for diffusion:

$$\bar{\mathbf{z}} = \frac{\mathbf{z} - \boldsymbol{\mu}_z}{\boldsymbol{\sigma}_z + \epsilon}. \quad (9)$$

3.2 FLOW MATCHING OVER GEOMETRY STATES

The flow model generates standardized compact geometry latents $\bar{\mathbf{z}}_{1:V}$, not RGB frames. Let $\bar{\mathbf{z}}_{1:V}^{\text{gt}}$ be the all-view target latent from the frozen codec and $\bar{\mathbf{z}}^{(1)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ be Gaussian noise. We use the linear path

$$\bar{\mathbf{z}}_t = (1 - t)\bar{\mathbf{z}}^{\text{gt}} + t\bar{\mathbf{z}}^{(1)}, \quad t \in [0, 1], \quad (10)$$

with velocity target $\mathbf{u}_t = \bar{\mathbf{z}}^{(1)} - \bar{\mathbf{z}}^{\text{gt}}$. A Diffusion Transformer $\mathbf{f}_\psi$ receives the noisy state, timestep, and conditioning signals and is trained by

$$\mathcal{L}_{\text{flow}} = \mathbb{E} \left[\rho(t) \left\| \hat{\mathbf{u}}_t - \left(\bar{\mathbf{z}}^{(1)} - \bar{\mathbf{z}}^{\text{gt}} \right) \right\|_2^2 \right], \quad (11)$$

where $\rho(t)$ is an optional time weight. The same scalar t is shared across all views in a training clip, so the model learns a coupled multi-view vector field rather than independent per-frame denoisers.

x -prediction for representation latents. Following RAEv2 (Singh et al., 2026), we parameterize the network to predict the clean latent $\hat{\bar{\mathbf{z}}}^{\text{gt}}$ and convert it to a velocity for the loss and sampler:

$$\hat{\mathbf{u}}_t = \frac{\bar{\mathbf{z}}_t - \hat{\bar{\mathbf{z}}}^{\text{gt}}}{\max(t, t_\epsilon)}, \quad t_\epsilon = 0.05. \quad (12)$$

This keeps the model output in the same coordinate system consumed by the decoder while avoiding unstable division near $t = 0$. Sampling uses Euler steps from $t = 1$ to $t = 0$, with optional classifier-free guidance.

Backbone. $\mathbf{f}_\psi$ is a DDT-style encoder-decoder transformer. Latent maps are patch-embedded with 1×1 patches, so each compact latent cell remains a token. Encoder blocks perform cross-view self-attention over all view tokens with camera-aware positional encoding; decoder blocks predict the output latent for the generated view slots. The temporal variant allocates part of the rotary position embedding to a frame axis, improving ordering when camera motion alone does not determine time.

3.3 REFERENCE, CAMERA, AND TEXT CONDITIONING

The main conditioning principle is that only generated views belong to the ODE state. Observed images, cameras, and text specify the state but do not themselves evolve. Table 1 summarizes this separation.

Table 1: **Roles of signals in NGD.** Generated views are noisy ODE states. Reference images are clean evidence tokens; cameras and text are controls.

Signal	Role	Enters as	ODE state?
Generated views	state	noisy latent tokens $\bar{\mathbf{z}}_t$	yes
Reference views	evidence	clean latent tokens $\bar{\mathbf{z}}^{\text{ref}}$	no
Camera path	geometric control	Plücker ray PE in attention	no
Text prompt	semantic control	cross-attention tokens	no

Train/inference-aligned reference tokens. Geometry encoders are set-based: the feature of a reference image encoded jointly with target views differs from the feature of the same image encoded alone. Placing a clean reference latent inside the flow state therefore creates a train/inference mismatch, because full-set reference features exist during training but not at test time. NGD instead encodes the K observed reference views in their inference-time context and prepends them as clean conditioning tokens:

$$\bar{\mathbf{z}}^{\text{ref}} = \text{Std} [\mathcal{A}_\phi(\mathcal{E}(\mathbf{I}_{1:K}^{\text{ref}}))] . \quad (13)$$

All V output slots, including slots at reference camera poses, remain noisy flow variables following Eq. 10. Reference tokens use timestep $t = 0$, attend with generated view tokens, and are dropped before the decoder prediction head. This yields a single standard Euler sampler and removes the denoising-level mismatch that arises when reference views are clamped inside the ODE.

Metric ray-space camera conditioning. For each latent cell and camera we compute a world-space ray in Plücker form $(\mathbf{d}, \mathbf{m})$, where $\mathbf{d}$ is direction and $\mathbf{m} = \mathbf{o} \times \mathbf{d}$ is the moment. We decompose it as

$$\mathbf{d}, \quad \hat{\mathbf{m}} = \frac{\mathbf{m}}{\|\mathbf{m}\| + \epsilon}, \quad s = \log(\|\mathbf{m}\| + \epsilon), \quad (14)$$

and inject the resulting ray embedding into query/key self-attention, following the ray-space conditioning philosophy of RayPE (Yin et al., 2026). The log-scale channel preserves metric translation magnitude when metric camera sidecars are available; for non-metric datasets we fall back to per-clip normalization for stability. Reference tokens receive the rays of their corresponding reference cameras, so the model knows where the clean evidence was observed.

Text conditioning. Text prompts are encoded by a frozen language model and projected to the transformer cross-attention width. Text dropout, reference dropout, and camera dropout expose the same weights to text-to-image, text-to-video, and reference-conditioned generation regimes.

3.4 TRAINING AND INFERENCE

Training proceeds in two stages. We first train the codec $\mathcal{A}_\phi$ on mixed single-view and multi-view data, selecting checkpoints using reconstruction quality together with the latent diagnostics in Section 4.1. We then freeze the codec, compute latent normalization statistics, and train the flow model over standardized compact latents with randomized view counts, reference counts, camera paths, and condition dropout.

At inference, NGD samples Gaussian noise for all V generated slots, encodes reference evidence if provided, integrates Eq. 12 from $t = 1$ to $t = 0$, denormalizes the final latents, and decodes them to RGB and geometry. Different tasks are obtained by masking conditions rather than changing the backbone:

Task	V	Text	Camera rays	Ref tokens K
Text-to-image	1	yes	–	0
Text-to-video / camera path	$V > 1$	yes	yes	0
Image/video-to-video	$V > 1$	optional	yes	≥ 1
Novel-view synthesis	$V > 1$	optional	yes	≥ 1

No hierarchy-level cascade or per-layer diffusion model is required: NGD’s compact latent is the sole generated state, and all outputs are decoded from it. Sections 4.1–4.3 evaluate whether this

design improves latent quality, reconstruction, and controllable generation relative to pixel VAE, semantic RAE, and raw geometry-feature baselines.

4 EXPERIMENTS

Our experiments are organized to separate three questions, matching our contributions: (i) is the compact geometry latent a better *generation* space than pixel-VAE, semantic-RAE, and raw geometry-feature latents, under a matched backbone and training budget (Section 4.1); (ii) does the compact autoencoder preserve reconstruction and fine-grained spatial detail relative to raw geometry features (Section 4.2); and (iii) do these latent-space gains translate into camera-controllable, geometry-consistent video and novel-view generation, with the latent’s effect isolated by holding the camera-injection mechanism fixed across all methods (Section 4.3).

4.1 LATENT-SPACE STUDY AND CONTROLLED LATENT ABLATION

Table 2 reports intrinsic latent diagnostics. No baseline latent is good on all axes: pixel VAEs are well-conditioned but have high VIV and weak structure, semantic RAEs are well-structured but do not decode geometry, and raw geometry features are geometry-native but high-dimensional and poorly conditioned. To expose the layer trade-off, we evaluate both the first DA3-GIANT out-layer (L0, the feature diffused by the matched DA3-Giant L0 baseline) and the deepest out-layer (L3). L0 preserves much stronger cross-view correspondence, whereas L3 posts lower raw VIV; however both layers have only about 10 effective channel dimensions out of 3072 and large condition numbers, so neither single raw layer is a broadly smooth generation state. The compact NGD latent instead gives a far lower-dimensional, better-conditioned state with much stronger single-view semantic/spatial structure. Table 3 probes the multi-view axis directly via cross-view correspondence retrieval. Raw DA3 level-0 features are strong here, as expected from a geometry encoder with cross-view attention; the challenge is turning that correspondence-bearing but ill-conditioned feature into a controllable generation state. This layer-wise behavior supports the autoencoder design in Section 3.1. Level 0 is valuable because it yields the best wide-baseline retrieval among the diagnostic rows ($\times\text{LNC}^* = 0.526$), but it is not by itself an attractive flow state (3072 channels, $\kappa \approx 10^8$, effective rank ≈ 10). Level 3 moves in the other direction: it has lower raw VIV (0.55 single-view, 0.34 multi-view) but much weaker single-view structure and cross-view retrieval, and an even worse single-view condition number (6.6×10^{16}). NGD therefore does not diffuse any single raw DA3 layer. It fuses levels 0–3: the shallow layer contributes correspondence, the deeper layers contribute the hierarchical geometry/semantic information needed for decoding, and the learned bottleneck repairs dimensionality and conditioning.

To test whether these diagnostics predict generation quality, we compare each latent under the same evaluation protocol (Table 4). The rows use the same flow-matching family and camera-conditioning protocol. The raw geometry-feature row is our framework-matched DA3-Giant L0 baseline: it diffuses the same DA3-GIANT level-0 feature evaluated in the diagnostics, then uses the frozen DA3 backbone to propagate deeper features for geometry decoding.

The raw L0/L3 comparison exposes both sides of geometry-foundation features. L0 is the best correspondence descriptor in Table 3, with $\times\text{LNC}^* = 0.642/0.526$ and centered $\times\text{LDS} = 0.518/0.386$ at near/wide baselines, whereas L3 drops to $\times\text{LNC}^* = 0.380/0.237$. L3, in turn, has lower raw VIV (0.55/0.34 vs. L0’s 0.86/0.75), but this coincides with weaker structure and severe spectral degeneracy rather than a uniformly better transport space. This is precisely the regime our multi-level VAE is designed for: retain the shallow correspondence signal, combine it with deeper DA3 geometry features, and expose the generator only to a compact, well-conditioned latent.

Extended 3D-consistency metrics. Table 6 reports the 3D-consistency metrics for the same latents as the appearance comparison in Table 4, under two reconstruction models: the independent VGGT estimator and a DA3-GIANT cross-check under a second reconstruction model. For every scene we run the reconstruction model on both the ground-truth clip and the generated clip (same 9 views), recover per-view cameras, depth, and point maps, and compare the generated geometry against the ground-truth clip’s under a Sim(3) (Umeyama) alignment. This “video-to-video” protocol needs no external pose labels and lets systematic estimator biases partly cancel. **ATE** and **RPEt/RPEr** are the RMSE absolute-trajectory error and relative-pose translation/rotation er-

Table 2: **Latent-space diagnostics.** Seven latents: pixel VAEs (SD-VAE, WAN2.1 VAE), a semantic RAE (RAEV2), raw DA3-GIANT L0/L3 features, and our compact NGD latents. VIV, its dimension-free companion ρ , covariance condition number κ , and effective rank summarize diffusability/spectrum; LNC/LDS/SRSS use uncentered single-view features ($\downarrow/\uparrow$ as marked). No single raw DA3 layer is good on all axes (L0: correspondence but ill-conditioned; L3: low raw VIV but weak structure and spectral degeneracy). Definitions and centered variants in Appendix A. Per column, the **best** and **second-best** across latents are shaded. NGD-64 is evaluated with the same DA3-GIANT $C=64$ checkpoint on both ImageNet-S and the multi-view geometry protocol.

Latent	$C \times H \times W$	Single-view (ImageNet)				Multi-view (geometry)			Structure (single-view)		
		VIV $\downarrow$	$\rho \downarrow$	$\kappa \downarrow$	effrank $\uparrow$	VIV $\downarrow$	$\rho \downarrow$	$\kappa \downarrow$	LNC@5 $\uparrow$	LDS $\uparrow$	SRSS $\uparrow$
Pixel VAE, single-image (SD-VAE)	$4 \times 32 \times 32$	5.03	0.888	3.7	3.0	3.08	0.765	4.1	0.051	0.262	0.324
Pixel VAE, video (WAN2.1 VAE)	$16 \times 32 \times 32$	1.77	0.847	127	5.3	0.96	0.609	160	0.084	0.234	0.313
Semantic RAE (RAEV2)	$1024 \times 18 \times 18$	0.87	0.803	5.1e13	239.7	0.80	0.726	3.6e13	0.931	0.365	0.519
Geometric VAE (DA3-Giant L0)	$3072 \times 18 \times 18$	0.86	0.876	2.8e8	11.3	0.75	0.793	3.9e8	0.353	0.102	0.123
Geometric VAE (DA3-Giant L3)	$3072 \times 18 \times 18$	0.55	0.769	6.6e16	11.5	0.34	0.539	2.0e11	0.054	0.054	0.070
Ours (NGD)	$128 \times 18 \times 18$	0.58	0.674	227	51.3	0.67	0.564	356	0.498	0.444	0.553
Ours (NGD-64 dim)	$64 \times 18 \times 18$	0.574	0.634	53.1	37.0	0.641	0.519	117	0.488	0.631	0.767

Table 3: **Cross-view correspondence consistency.** $\times\text{LNC}^*$ is chance-normalized cross-view retrieval (0=random, 1=perfect) via off-the-shelf UFM matches, testing whether corresponding points map to nearby latent tokens. Shallow L0 retains far stronger correspondence than deep L3, yet Table 2 shows neither raw layer alone is a good generation state. (UFM uses a DINOv2 backbone, biasing RAEV2 upward.) Per column, the **best** / **second-best** across latents are shaded. The NGD-64 row is the completed DA3-GIANT $C=64$ run.

Latent	$\times\text{LNC}^*$ near $\uparrow$	$\times\text{LNC}^*$ wide $\uparrow$
Pixel VAE, single-image (SD-VAE)	0.124	0.079
Pixel VAE, video (WAN2.1 VAE)	0.402	0.260
Semantic RAE (RAEV2)	0.458	0.333
Geometric VAE (DA3-Giant L0)	0.642	0.526
Geometric VAE (DA3-Giant L3)	0.380	0.237
Ours (NGD)	0.591	0.405
Ours (NGD-64 dim)	0.569	0.381

ror (degrees); **MEt3R** is the official multi-view feature-consistency score, computed on the generated views only and therefore reconstruction-model-agnostic (one value per latent). We report two estimators: VGGT shares no backbone with any evaluated latent (the independent estimator used above), while DA3-GIANT is a stronger but *not* fully independent probe for the DA3-derived latents (DA3-Giant L0/L3 and NGD) and is included only as a robustness cross-check, comparable by *ranking* rather than absolute scale. Under both estimators and on both datasets the ordering is stable: the compact NGD latent is the most consistent among the seven latent rows on VGGT trajectory/reprojection and MEt3R (and on most DA3-GIANT columns), while the raw single-layer geometry features (DA3-Giant L0/L3) remain weak despite being geometry-native. Figure 4 plots five VGGT consistency metrics as a dedicated radar chart, where NGD spans a full, balanced polygon while every baseline collapses on at least one axis; the same information is aggregated into the single 3D-consistency axis of the latent-quality radar (Figure 1).

Qualitative comparison. Figures 5 and 6 show the same latents on shared scenes from both datasets, sampled from the matched 64-scene pools with one reference view and identical camera conditioning, so every column is the same target viewpoint. The failure modes follow the metrics. The video pixel VAE (WAN2.1) and the deep raw geometry feature (DA3-Giant L3) lose object structure and drift in colour as the camera moves away from the reference; RAEV2 keeps global layout but softens high-frequency detail; DA3-Giant L0 stays sharp near the reference but degrades at wide baselines, consistent with its weak trajectory metrics in Table 6. Both NGD latents keep object identity and scene layout stable across the full trajectory.

Table 4: **Latent comparison on RealEstate10K and DL3DV under the same protocol (appearance)**. The two datasets are shown side by side; appearance metrics are measured on the generated clips. Both use the matched RealEstate10K+DL3DV finetunes (64 scenes $\times$ 9 views at 252², same scene pool as Table 9); WAN2.1 is the RE10K-trained temporal WAN baseline (no *_dl3dv finetune), evaluated on both datasets. Geometry/3D-consistency for the same models (VGGT and DA3-GIANT estimators, Met3R) is reported separately in Table 6. The **best** and **second-best** among the seven latent rows are shaded (darker blue = best); the **gray** GLD (Jang et al., 2026) and Gen3R (Huang et al., 2026) rows are external comparison systems and are excluded from the ranking; see also Table 9.

Latent	RealEstate10K					DL3DV				
	FVD↓	FID↓	LPIPS↓	PSNR↑	SSIM↑	FVD↓	FID↓	LPIPS↓	PSNR↑	SSIM↑
Pixel VAE, single-image (SD-VAE)	258.6	36.4	0.171	19.32	0.666	373.2	57.1	0.232	17.05	0.495
Pixel VAE, video (WAN2.1 VAE)	362.9	41.0	0.219	16.52	0.575	596.7	77.3	0.313	14.31	0.407
Semantic RAE (RAEV2)	379.4	31.4	0.191	17.60	0.585	453.4	49.2	0.231	16.01	0.434
Geometric VAE (DA3-Giant L0)	298.6	35.5	0.177	18.31	0.643	376.5	60.2	0.235	16.68	0.481
Geometric VAE (DA3-Giant L3)	488.9	50.0	0.273	16.82	0.579	584.8	80.6	0.314	16.22	0.447
Ours (NGD)	233.4	25.6	0.146	19.54	0.701	345.2	44.2	0.200	17.39	0.541
Ours (NGD-64 dim)	225.7	27.1	0.143	20.02	0.711	287.0	41.3	0.194	18.00	0.553
GLD (Jang et al., 2026)	445.1	44.0	0.337	14.91	0.518	587.4	66.4	0.356	14.51	0.406
Gen3R (Huang et al., 2026) (WAN-pretr.)	269.7	25.7	0.196	17.79	0.621	580.5	48.3	0.270	15.62	0.440

Table 5: **Inference timing and parameter summary**. Per-scene times for 9 views, one reference view, 50 Euler steps, CFG = 2, and BF16. Times are CUDA-synchronized and exclude model/data loading. VAE time combines the separate latent/VAE decoder (when present) and one RGB-plus-geometry output path, excluding metrics and artifact writing. **DiT** counts the diffusion transformer; **VAE** counts the learned RGB/geometry output modules. For raw DA3-Giant L0, this includes its RGB decoder, DA3 propagation blocks, and DPT geometry head.

Method	Time (s)			Parameters (B)	
	Diffusion	VAE	Total	DiT	VAE
<i>RGB+geometry output</i>					
Gen3R (Huang et al., 2026) (WAN-pretr.)	85.706	2.105	87.811	1.616	0.517
GLD (Jang et al., 2026)	10.774	0.127	10.901	1.256	0.438
Geometric VAE (DA3-Giant L0)	10.531	0.294	10.825	1.052	0.823
Ours (NGD)	10.045	0.451	10.496	0.935	0.327
<i>RGB-only output</i>					
Pixel VAE, single-image (SD-VAE)	10.027	0.043	10.070	0.933	0.049
Pixel VAE, video (WAN2.1 VAE)	9.181	0.061	9.242	0.934	0.073
Semantic RAE (RAEV2)	9.452	0.175	9.627	0.956	0.416
Geometric VAE (DA3-Giant L3)	9.959	0.017	9.976	1.052	0.206

Cross-dataset transfer. Table 4 reports appearance for the same latent families on held-out DL3DV under the matched protocol, and Table 6 their 3D-consistency. NGD remains best among the latent rows on appearance and every VGGT consistency axis out of domain (ATE 0.0056 vs. 0.0096 for Gen3R; RPEr 0.294° vs. 0.678°; Met3R 0.1347 vs. 0.1662; Table 6), and also leads frame-wise FID on the 64-scene pool — mirroring the RealEstate10K ordering.

Compact codec ablations. Holding the DiT recipe fixed, we ablate the codec backbone, latent width ($C=64$ vs. $C=128$), and the representation-alignment term $\mathcal{L}_{\text{repr}}$ (dual REPA) used to train the autoencoder (Table 7). These are codec-design axes under one matched generation protocol, distinct from the conditioning ablations of Section 4.3, which change the flow objective. The shifted DA3-GIANT $C=64$ codec gives the strongest visual generation metrics among these variants, while $C=128$ retains a slight edge on camera and multi-view consistency metrics. With DA3-LARGE, removing $\mathcal{L}_{\text{repr}}$ improves FVD but consistently weakens paired-frame and geometry quality, supporting its role in preserving semantic and spatial structure beyond raw perceptual fidelity.

Compute and data efficiency. A compact, well-conditioned, structured latent should also be more economical to learn in. In a matched-resource study, we will compare quality as a function of

Table 6: **Extended 3D-consistency metrics on RealEstate10K and DL3DV (video-to-video; lower is better)**. Two reconstruction models recover cameras/geometry from generated vs. ground-truth clips. ATE/RPE are Sim(3)-aligned; Reproj. is the 50% confidence reprojection score. VGGT is backbone-independent of all latents; DA3-GIANT is a cross-check, comparable by ranking only. MEt3R is recon-model-agnostic. Both blocks use the matched RealEstate10K+DL3DV finetunes (64 scenes $\times$ 9 views, same pool as Table 9); WAN2.1 is the RE10K-trained temporal WAN baseline (no *_dl3dv finetune), evaluated on both datasets. The **best** and **second-best** among the seven latent rows are shaded (darker blue = best); the **gray** GLD (Jang et al., 2026) and Gen3R (Huang et al., 2026) rows are external comparison systems and are excluded from the ranking.

Latent	VGGT (independent)				DA3-GIANT (cross-check)				
	ATE $\downarrow$	RPE $\downarrow$	RPE $^{\circ}\downarrow$	Reproj. $\downarrow$	ATE $\downarrow$	RPE $\downarrow$	RPE $^{\circ}\downarrow$	Reproj. $\downarrow$	MEt3R $\downarrow$
<i>RealEstate10K</i>									
Pixel VAE, single-image (SD-VAE)	0.0072	0.0182	0.648	0.0093	0.0046	0.0191	0.338	0.0018	0.1385
Pixel VAE, video (WAN2.1 VAE)	0.0225	0.0252	0.746	0.0105	0.0135	0.0176	0.751	0.0019	0.1441
Semantic RAE (RAEV2)	0.0085	0.0233	0.420	0.0104	0.0066	0.0219	0.507	0.0019	0.1263
Geometric VAE (DA3-Giant L0)	0.0085	0.0202	0.435	0.0096	0.0047	0.0167	0.358	0.0020	0.1420
Geometric VAE (DA3-Giant L3)	0.0112	0.0211	0.516	0.0124	0.0036	0.0148	0.355	0.0019	0.1575
Ours (NGD)	0.0041	0.0173	0.332	0.0075	0.0026	0.0143	0.304	0.0017	0.1213
Ours (NGD-64 dim)	0.0034	0.0060	0.380	0.0068	0.0026	0.0141	0.225	0.0013	0.1208
GLD (Jang et al., 2026)	0.0124	0.0258	0.886	0.0133	0.0088	0.0221	0.950	0.0038	0.1660
Gen3R (Huang et al., 2026) (WAN-pretr.)	0.0090	0.0213	0.423	0.0072	0.0063	0.0211	0.387	0.0014	0.1157
<i>DL3DV</i>									
Pixel VAE, single-image (SD-VAE)	0.0073	0.0098	0.432	0.0108	0.0055	0.0076	0.358	0.0024	0.1542
Pixel VAE, video (WAN2.1 VAE)	0.0243	0.0288	1.419	0.0258	0.0206	0.0251	1.016	0.0019	0.1777
Semantic RAE (RAEV2)	0.0104	0.0143	0.638	0.0130	0.0088	0.0113	0.585	0.0031	0.1549
Geometric VAE (DA3-Giant L0)	0.0110	0.0136	0.634	0.0129	0.0079	0.0102	0.482	0.0042	0.1721
Geometric VAE (DA3-Giant L3)	0.0097	0.0120	0.603	0.0151	0.0052	0.0073	0.380	0.0039	0.1754
Ours (NGD)	0.0056	0.0072	0.294	0.0080	0.0043	0.0053	0.260	0.0029	0.1347
Ours (NGD-64 dim)	0.0056	0.0066	0.293	0.0077	0.0038	0.0053	0.254	0.0028	0.1370
GLD (Jang et al., 2026)	0.0136	0.0164	1.242	0.0241	0.0104	0.0136	1.048	0.0087	0.2183
Gen3R (Huang et al., 2026) (WAN-pretr.)	0.0096	0.0133	0.678	0.0143	0.0096	0.0131	0.608	0.0065	0.1662

Table 7: **Compact codec ablations (DiT recipe fixed)**. C is the VAE latent channel width; $\mathcal{L}_{\text{repr}}$ is the dual REPA alignment term in the autoencoder objective. Results use matched 64-scene, 9-view pools on RealEstate10K and DL3DV; each variant uses its latest DL3DV-finetuned checkpoint, with the DA3-GIANT $C=64$ row using the latest ordinary `giant64` variant. PSNR is a paired-frame metric; ATE is from the independent VGGT estimator, and MEt3R is the official multi-view consistency metric. Per column, the **best / second-best** within each dataset are shaded.

VAE encoder	C	$\mathcal{L}_{\text{repr}}$	<i>RealEstate10K</i>				<i>DL3DV</i>			
			FVD $\downarrow$	PSNR $\uparrow$	ATE $\downarrow$	MEt3R $\downarrow$	FVD $\downarrow$	PSNR $\uparrow$	ATE $\downarrow$	MEt3R $\downarrow$
DA3-GIANT	64	yes	225.7	20.02	0.0034	0.1208	287.0	18.00	0.0056	0.1370
DA3-GIANT	128	yes	244.4	19.66	0.0032	0.1214	345.9	17.28	0.0055	0.1356
DA3-LARGE	128	yes	266.4	19.19	0.0044	0.1192	388.4	16.76	0.0066	0.1357
DA3-LARGE	128	no	242.5	19.20	0.0033	0.1226	333.0	17.07	0.0058	0.1374

compute or data budget against both a pixel VAE and the matched raw DA3-GIANT level-0 feature baseline (Figure 7). We expect the compact latent to be more efficient than raw geometry features because it is far lower-dimensional (128 vs. 3072 channels) and far better conditioned ($\kappa \approx 10^2$ vs. 10^8 ; Table 2). All comparisons are reproduced within our own framework rather than taken across papers.

4.2 RECONSTRUCTION FIDELITY OF THE COMPACT LATENT

A compact latent risks discarding the fine-grained spatial detail that raw geometry features retain. Table 8 therefore separates reconstruction from generation-state quality. A separately trained RGB head on raw DA3 L0 gives the strongest distortion/perceptual reconstruction, as expected for a high-dimensional shallow feature that preserves local appearance. However, Tables 2–3 show that this same feature remains a poor diffusion state: it is 3072-channel, ill-conditioned, and nearly rank-

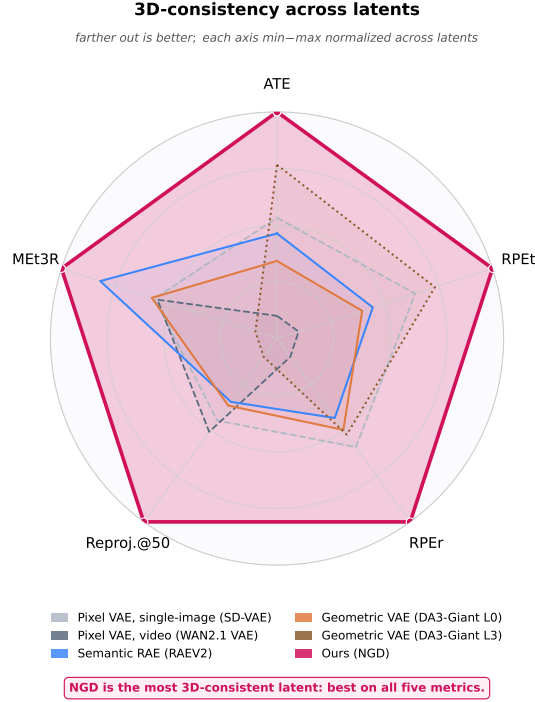


Figure 4: **3D-consistency radar.** Five VGGT metrics (ATE, RPET, RPER, 50% reprojection, MET3R), min-max normalized across six latents and oriented so farther out is better. NGD leads on all five; every baseline collapses on at least one — pixel VAEs on reprojection, raw geometry features on the trajectory metrics.

Table 8: **Reconstruction.** Single-view = ImageNet val at 256^2 (sample rFID over 2000 images); multi-view = geometry-scene reconstruction (rFVD with PSNR/LPIPS over clips). Raw DA3 L0/L3 use separately trained RGB heads on the frozen feature level. L0 reconstructs RGB best but is not a good diffusion state (Tables 2–3); NGD nearly matches it while compressing the DA3 stack to a compact, better-conditioned latent. Per column, the **best / second-best** across latents are shaded.

Latent	Single-view (ImageNet)			Multi-view (geometry)		
	PSNR↑	LPIPS↓	rFID↓	PSNR↑	LPIPS↓	rFVD↓
Pixel VAE, single-image (SD-VAE)	18.49	0.140	10.1	19.97	0.121	37.7
Pixel VAE, video (WAN2.1 VAE)	18.34	0.125	8.9	19.83	0.163	46.7
Semantic RAE (RAEV2)	22.08	0.134	8.8	24.73	0.094	32.1
Geometric VAE (DA3-Giant L0)	28.64	0.038	5.5	34.70	0.014	4.4
Geometric VAE (DA3-Giant L3)	19.90	0.219	25.3	23.98	0.122	76.2
Ours (NGD)	28.76	0.036	5.4	34.78	0.014	4.3
Ours (NGD-64 dim)	27.30	0.061	7.1	32.37	0.040	9.1

degenerate. Raw L3 moves in the opposite direction and loses much of the invertible appearance information, producing substantially worse RGB reconstruction. NGD nearly matches raw L0 across PSNR, LPIPS, and sample rFID while compressing the DA3 stack to a compact, better-conditioned latent suitable for flow matching.

Native geometry decoding (no degradation). A compact latent could still be a good *RGB* state while quietly losing the geometry it claims to carry. Table 9 rules this out by scoring the geometry NGD decodes *natively* — per-view depth, cameras from the ray head, and back-projected point maps — against a neutral $Pi3$ reconstruction of the *real* target frames at the same viewpoints (64

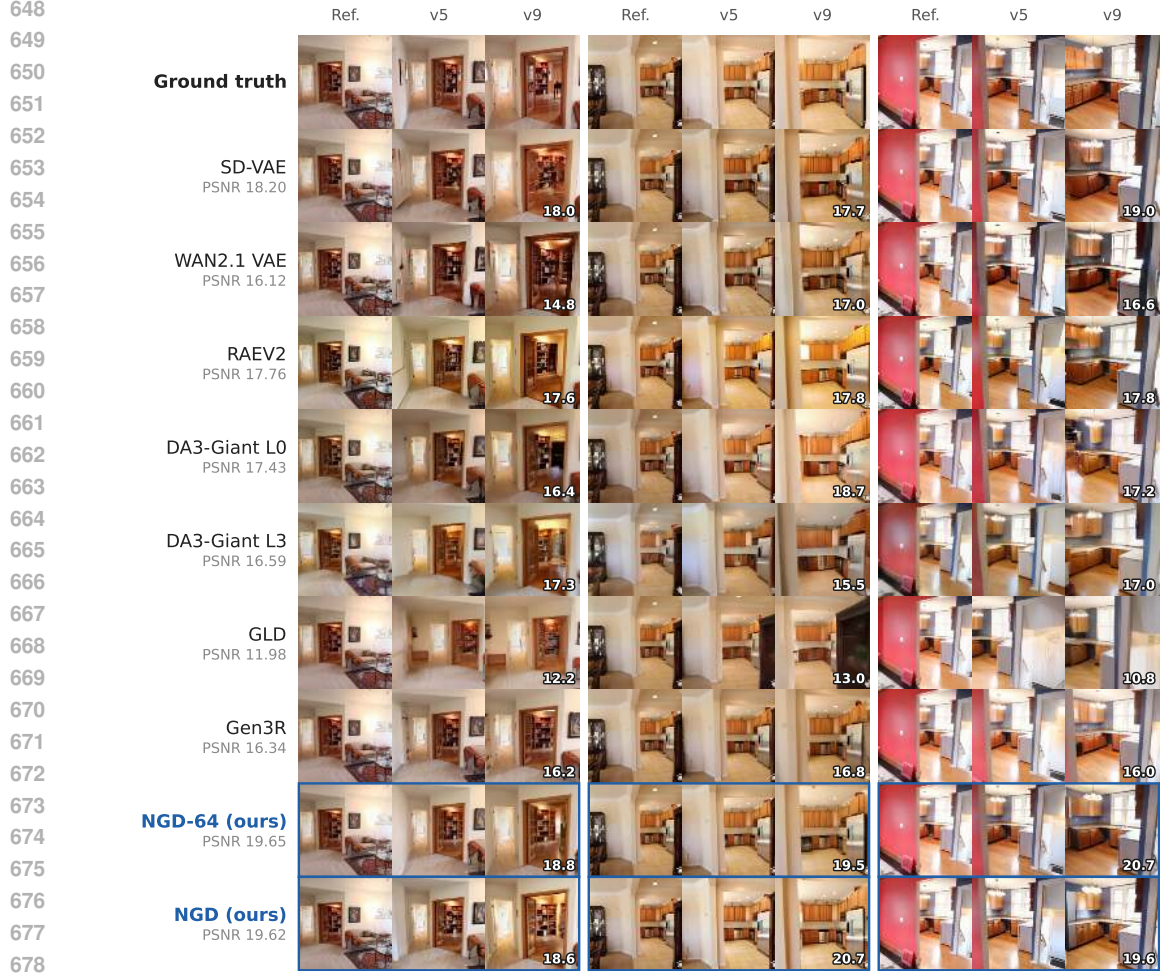


Figure 5: **Qualitative comparison on RealEstate10K.** Rows: the seven latents plus GLD and Gen3R. Each block is one scene — a reference view and two targets (5, 9) under identical camera conditioning. Inset numbers are per-scene PSNR over the eight generated views; the grey number under each label is the mean over the three scenes. NGD (blue) keeps structure and layout at wide baselines, while WAN2.1 and DA3-Giant L3 lose structure and DA3-Giant L0 drifts from the reference. Scenes are high-motion clips where both NGD rows beat every baseline (Appendix B).

scenes $\times$ 9 views on RealEstate10K and held-out DL3DV). Depth and point maps are reported as agreement with that reconstruction (median-scale depth; Chamfer/PMap under *pose-mediated* Sim(3) from camera trajectories); poses (ATE/RPer) are scored against the *real* dataset trajectory and are estimator-independent. Crucially, we score generated geometry against Pi3 on the *real* frames rather than on each model’s own frames: the latter is circular and not comparable across generators. We evaluate two states — encode $\rightarrow$ decode reconstruction on real frames (NGD/L0 VAE-direct) and camera-conditioned generation — against three baselines: the framework-matched raw DA3-Giant L0 feature latent; concurrent GLD (Jang et al., 2026), which samples a shallow DA3 layer and cascades to L0; and Gen3R (Huang et al., 2026), which uses two aligned latents (WAN+VGGT) rather than a single geometry-native state. On RealEstate10K, NGD’s VAE-direct reconstruction is far ahead of raw L0 on every column (0.090 vs. 0.178 AbsRel; 0.006 vs. 0.068 ATE) and leads Gen3R and GLD on pose, while Gen3R slightly edges AbsRel on depth alone. Under generation NGD remains the strongest DA3-native latent on depth and pose (0.139 AbsRel, 0.013 ATE); raw L0 is second on depth and point maps and slightly edges RPer, but GLD lags on all axes and Gen3R again degrades more from reconstruction to generation than NGD does. The same protocol on DL3DV preserves the ranking: NGD leads raw L0 in both regimes (recon. 0.125 vs. 0.234 AbsRel, 0.008 vs. 0.057 ATE; gen. 0.225 vs. 0.245 AbsRel, 0.325 vs. 0.330 Chamfer, 0.016

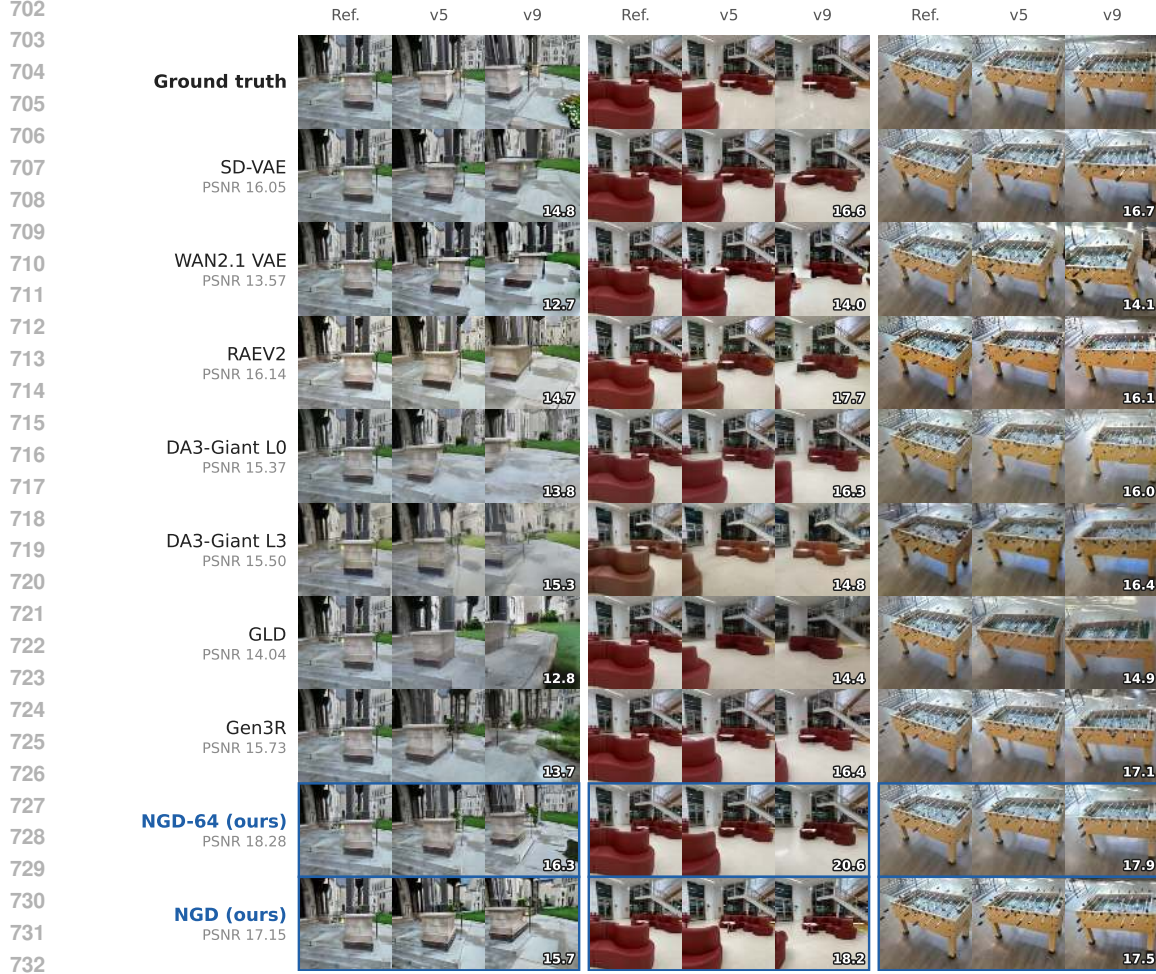


Figure 6: **Qualitative comparison on held-out DL3DV.** Same layout as Figure 5, on a dataset no latent was finetuned for. The NGD-to-baseline gap widens here (cf. Table 4). Full per-scene grids: Figures 12–21.

vs. 0.019 ATE), with GLD well behind and Gen3R showing the largest reconstruction→generation drop. Using Pi3 — independent of the DA3-derived latents and of Gen3R’s VGGT backbone — so the ordering is not a home-field artifact of either foundation model.

Figure 8 shows the corresponding qualitative behaviour on all three geometric outputs at once, against the same Pi3-on-real reference the table uses. Raw DA3-Giant L0 loses its depth scale at wide baselines, Gen3R collapses the far field into a hole and GLD saturates it against the near clip, while NGD’s natively decoded depth stays smooth and keeps the object boundaries Pi3 resolves. The camera-path column is the estimator-independent part of the comparison: NGD tracks the real trajectory as closely as Pi3 does from the real images (0.009 vs. 0.012 ATE on this scene), whereas all three baselines leave it once the camera turns (0.040, 0.042 and 0.069).

Sampling efficiency. Table 5 aggregates all evaluated models separately from the quality tables and identifies the timing scope for each row. Under the matched primary-output scope, NGD takes 10.496 s per scene, slightly faster than raw L0 at 10.825 s, while jointly producing RGB and geometry. The component breakdown also shows that decoding is a small fraction of total runtime for both methods (0.451 s for NGD and 0.294 s for raw L0); the dominant cost is diffusion.

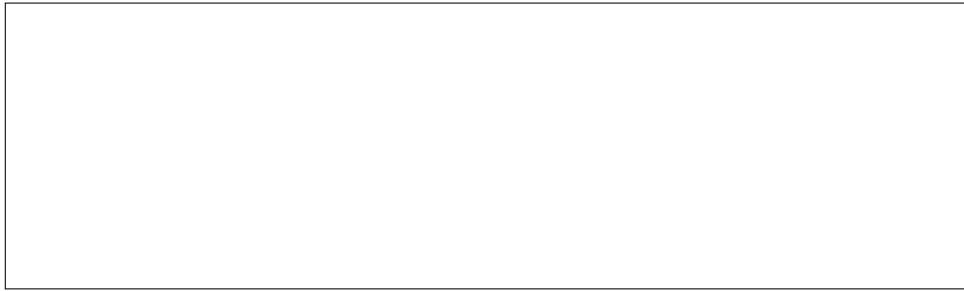


Figure 7: **TODO: Compute / data-efficiency curves.** Generation quality (FVD/FID) and geometry consistency (reprojection / point-cloud) vs. compute or data budget, comparing the NGD compact geometry latent against a pixel-VAE latent and the raw DA3-GIANT level-0 feature latent (DA3-Giant L0), all under a matched backbone and evaluation protocol within our own framework. Tests whether the compact latent reaches better quality for the same resource budget than both a pixel VAE and raw geometry features.

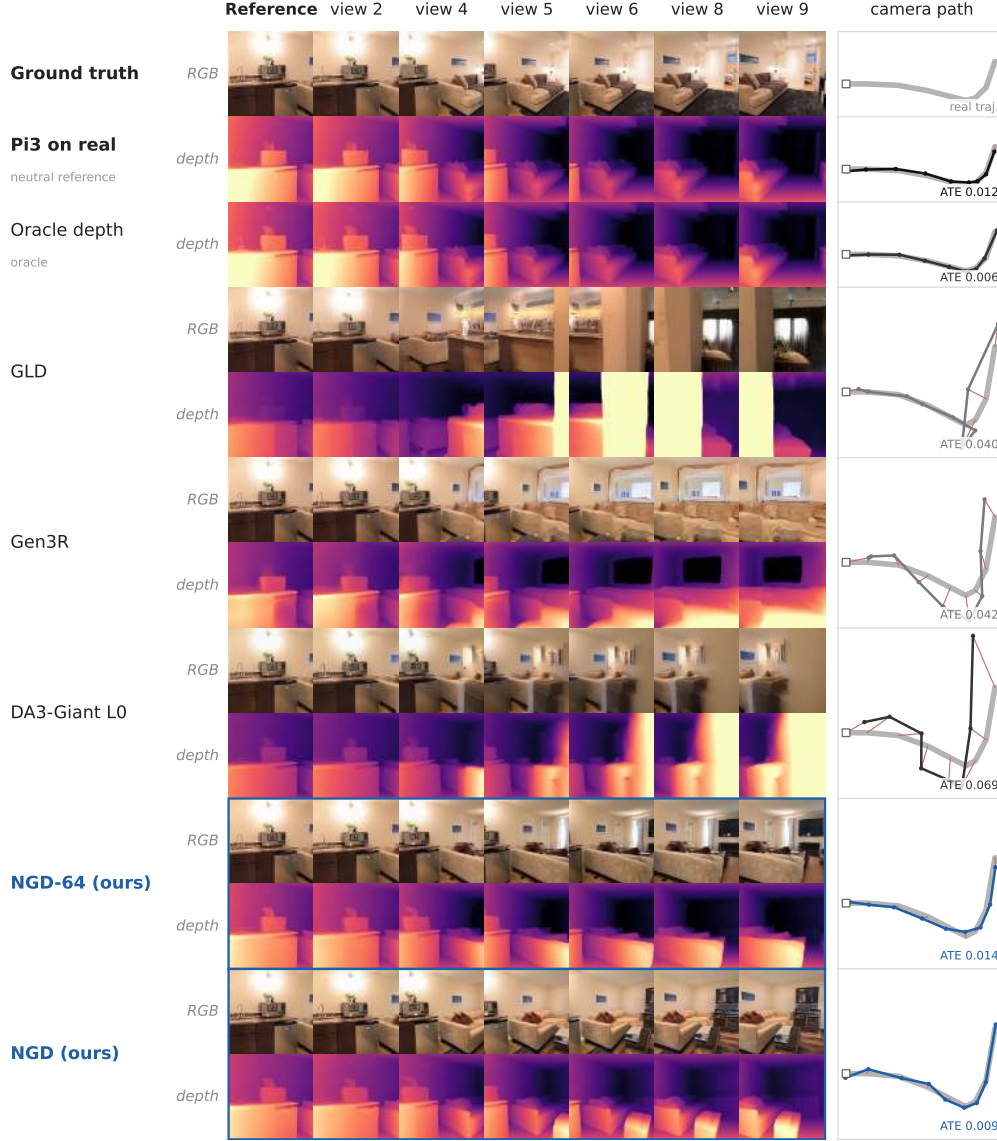


Figure 8: **Jointly decoded RGB, geometry and camera path.** One RealEstate10K scene; each method shows an RGB row and the depth decoded from the *same* state, under identical camera conditioning. Pi3 on real (Pi3 on the real frames) is the neutral reference used in Table 9; Oracle depth is our head on real-frame features — an upper bound, not neutral, since the NGD rows share it. Depth is inverse depth on one colour map after median scale alignment to Pi3, so colour differences are geometry errors, not scale. Last column: each method’s recovered camera centres (Sim(3)-aligned) over the real trajectory (grey), red spokes are per-frame residuals, cross-track axis magnified. Per-scene and DL3DV versions: Figures 22–25.

Table 9: **Native geometry decoding on RealEstate10K and DL3DV (no degradation).** The two datasets are shown side by side. Directly-decoded geometry vs. Pi3 run on the *real* target frames (64 scenes $\times$ 9 views; non-circular across rows). Columns: depth (AbsRel/ δ_1) and point maps (Chamfer/PMap-L1) as agreement with Pi3 (median-scale / pose-mediated Sim(3)); poses (ATE/RPEr) vs. the real trajectory. DA3-Giant L0 is the geometric-VAE baseline; GLD (Jang et al., 2026) is the concurrent L1+cascade sampler (same pipeline in both blocks; no VAE-direct); Gen3R (Huang et al., 2026) uses two aligned latents (WAN+VGGT), subsampled to the shared $252^2/9$ -view grid. Per column, the **best / second-best** among {GLD, DA3-Giant L0, NGD, NGD-64 dim} are shaded (the Pi3-on-real reference row is excluded from ranking); the **gray** Gen3R row is WAN-pretrained on large-scale data and is *not* a fair comparison. Lower is better except δ_1 .

Method	RealEstate10K (Pi3 ref)						DL3DV (Pi3 ref)					
	AbsRel $\downarrow$	$\delta_1 \uparrow$	Cham. $\downarrow$	PMap $\downarrow$	ATE $\downarrow$	RPEr $^\circ \downarrow$	AbsRel $\downarrow$	$\delta_1 \uparrow$	Cham. $\downarrow$	PMap $\downarrow$	ATE $\downarrow$	RPEr $^\circ \downarrow$
Pi3 on real (ref)	—	—	—	—	0.009	0.17	—	—	—	—	0.011	0.26
<i>Reconstruction (encode $\rightarrow$ decode; real frames)</i>												
Gen3R (Huang et al., 2026) (WAN-pretr.)	0.085	0.906	0.430	0.829	0.038	0.63	0.114	0.862	0.237	0.542	0.022	0.72
GLD (Jang et al., 2026)	0.271	0.694	0.857	1.580	0.039	1.23	0.303	0.608	0.462	1.083	0.026	1.14
Geometric VAE (DA3-Giant L0)	0.178	0.759	0.635	1.160	0.068	1.33	0.234	0.639	0.575	1.130	0.057	1.09
Ours (NGD)	0.090	0.905	0.400	0.700	0.006	0.19	0.125	0.862	0.145	0.433	0.008	0.25
Ours (NGD-64 dim)	0.090	0.905	0.386	0.691	0.007	0.20	0.125	0.861	0.145	0.433	0.008	0.27
<i>Generation (vs. Pi3 on the real target frames)</i>												
Gen3R (Huang et al., 2026) (WAN-pretr.)	0.164	0.807	0.727	1.428	0.046	0.69	0.232	0.688	0.587	1.180	0.031	0.91
GLD (Jang et al., 2026)	0.271	0.694	0.858	1.580	0.039	1.23	0.323	0.594	0.462	1.097	0.028	1.19
Geometric VAE (DA3-Giant L0)	0.157	0.822	0.540	1.058	0.019	0.35	0.245	0.715	0.330	0.815	0.019	0.48
Ours (NGD)	0.139	0.849	0.432	0.813	0.013	0.37	0.225	0.750	0.325	0.832	0.016	0.40
Ours (NGD-64 dim)	0.134	0.857	0.501	0.924	0.010	0.33	0.204	0.765	0.262	0.698	0.014	0.36

4.3 CONTROLLABLE VIDEO AND NOVEL-VIEW GENERATION

We finally evaluate NGD on camera-conditioned video and reference-conditioned novel-view synthesis, reporting both visual metrics (FVD, frame-FID, LPIPS/PSNR/SSIM on non-context frames) and geometry-aware metrics (reprojection and point-cloud consistency) with an independent estimator. Because every latent in the controlled comparison (Tables 4–6) already shares one fixed camera-injection mechanism, the cross-view consistency measured there is attributable to the latent rather than to pose conditioning, so we do not separately ablate the injection mechanism. Three conditioning/training ablations are summarized together in Table 10 (codec-width / $\mathcal{L}_{\text{repr}}$ ablations remain in Table 7).

Reference conditioning. We test whether the set-dependence of the geometry encoder actually produces the reference-target gap our formulation is designed to remove. Reference-in-state places the clean reference latent (computed from the full ground-truth view set at training time) into the flow state; our clean-token formulation keeps all views noisy and injects the reference only as a conditioning token. Key is the bidirectional Chamfer distance between DPT-decoded point clouds of reference vs. target slots. Clean tokens cut this gap by about $3\times$ while also slightly improving MET3R (Table 10).

Metric vs. normalized pose. We compare conditioning on metric camera pose against the per-scene normalized pose common in camera-conditioned generation. Normalized pose discards the metric translation scale; the Key column is VGGT SE(3) ATE *without* Sim(3) rescale, which drops from 0.0346 to 0.0093 under metric pose, while MET3R and FVD both improve (Table 10).

Text-to-image co-training. We co-train a BLIP3o-style text-to-image stream every third step ($\approx 33\%$) and ablate by disabling it. Co-training improves appearance and multi-view consistency (FVD 472.9 $\rightarrow$ 241.2; Key = LPIPS 0.199 $\rightarrow$ 0.147; Table 10). Beyond its role as an auxiliary training stream, the same checkpoint retains direct text-conditioned RGB and geometry generation. Figure 9 pairs each generated RGB image with its decoded depth and an expanded prompt caption, spanning photographic, illustrative, fantastical, and prompt-edited content. We include these samples as a qualitative capability check rather than a standalone text-to-image benchmark.

Table 10: **Generation conditioning ablations (NGD latent / backbone fixed)**. Each block changes one choice. Shared metrics are FVD, PSNR, and official MEt3R. Block-specific Key ($\downarrow$): Ref-tgt PC gap (median-scaled bidirectional Chamfer); SE(3) Pose-follow ATE without Sim(3) scale; LPIPS (T2I). Per column within each block, the **best** setting is shaded.

Ablation	Setting	FVD $\downarrow$	PSNR $\uparrow$	MEt3R $\downarrow$	Key $\downarrow$
<i>Reference conditioning (Key = Ref-tgt PC gap)</i>					
	Reference-in-state (V3)	257.1	18.91	0.1217	0.1016
	Clean conditioning token (ours)	241.2	19.47	0.1208	0.0318
<i>Pose scale (Key = SE(3) ATE, no Sim(3) scale)</i>					
	Per-scene normalized pose	350.7	16.28	0.1290	0.0346
	Metric pose (ours)	241.2	19.47	0.1208	0.0093
<i>T2I co-training (Key = LPIPS)</i>					
	w/o T2I co-train	472.9	16.77	0.1262	0.199
	w/ T2I co-train (ours)	241.2	19.47	0.1208	0.147

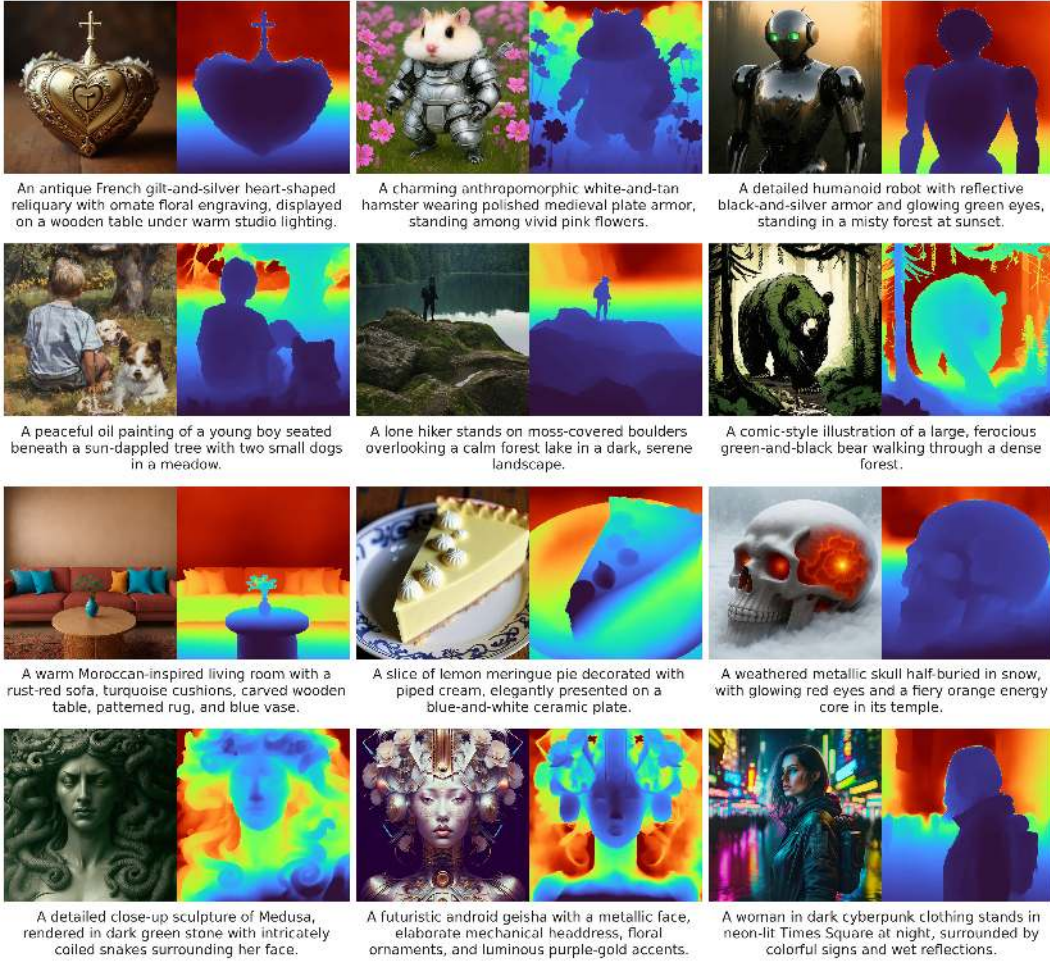


Figure 9: **Selected text-conditioned RGB and geometry generations**. For each prompt, the left image is the generated RGB output and the right image is its paired decoded depth. The expanded sentence below each pair records the subject, scene, style, and distinguishing prompt attributes. We select 12 representative examples from the larger qualitative set to balance visual and geometric readability with content diversity, covering realistic scenes and objects, oil painting, comics, science fiction, and fantastical portraiture. These examples illustrate the breadth retained by T2I co-training and are not a quantitative benchmark.

Long-rollout show cases. The comparisons above use nine views so that every latent is measured under one matched protocol. Figure 10 instead shows what a single decode of a longer rollout looks like: 33 views at 504^2 sampled from one reference view with 50 Euler steps and $\text{CFG} = 2$, then decoded once into RGB and into geometry. Each point cloud is the per-view depth of all 33 views unprojected with the camera read off the decoded rays and simply concatenated — no fusion across views, no external reconstruction and no test-time optimization — so whatever agreement is visible between views is agreement the latent already carried. Decoding a *sampled* latent this far out does need one extra step, because the geometry path is trained on encoded latents: we fine-tune the decoder (geometry adapter, decoder convolutions, shared trunk and RGB head, the latter with an RGB self-distillation term) for 6k steps on cached sampled latents from five domains against the frozen geometry decode of the re-encoded latent, leaving the DA3 encoder, the DPT head and the DiT untouched. These panels are selected cases and are meant as a show case rather than a measurement; Appendix C states how they were picked, what was left out and why, and shows each scene on its own page.

Qualitative observation: ray-addressed content reuse. As a minor side observation, because scene content is addressed in ray space, revisiting a camera ray can regenerate consistent content (Figure 11). We report this only qualitatively and do not claim it as a quantitative capability.

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A LATENT-SPACE DIAGNOSTICS: DEFINITIONS AND EXTENDED ANALYSIS

This appendix expands Table 2: it gives the definitions of every diagnostic we report, the complete per-encoder numbers (single-view ImageNet, resolution 256, 100 classes, 16 frames/class), and the extended analysis showing that the record-low VIV of raw geometry features is a dimensionality / dead-channel artifact.

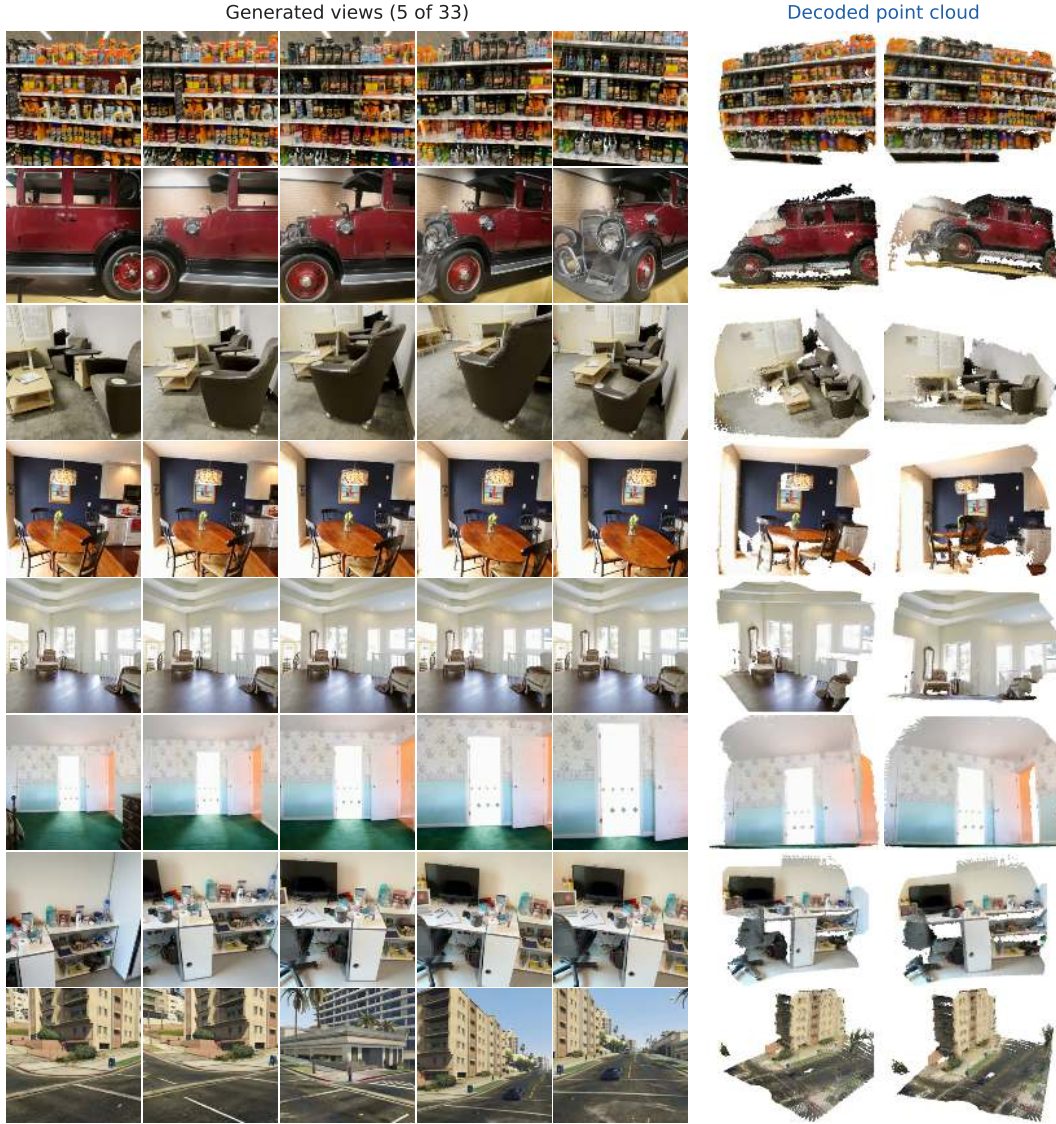


Figure 10: **Long-rollout show case: 33-view generation and the point cloud decoded from the same latents.** Eight scenes from the four domains the decoder was fine-tuned on. Rows, top to bottom: DL3DV supermarket, vintage car, lounge chairs; RealEstate10K dining room, bedroom, papered room; ScanNet++ office desk; MVS-Synth city block. *Left:* five of the 33 generated views, evenly spaced. *Right:* the geometry decode of the same latents from two orbit angles — per-view depth of all 33 views, unprojected with the camera recovered from the decoded rays, coloured by RGB and concatenated with no fusion. Angles are per scene and the far tail is trimmed; the MVS-Synth street additionally drops sky-coloured points and is cropped along the street axis (Appendix C). Nothing else is post-processed.

A.1 METRIC DEFINITIONS

Diffusability (transport smoothness under flow matching).

- **VIV** (velocity-irreducible variance): $VIV = \frac{\pi}{2} \mathbb{E}[\sqrt{\lambda}]$, the mean over the latent’s per-mode velocity spectrum $\{\lambda\}$; lower means smoother flow-matching transport. Because the expectation runs over *all* $C \times H \times W$ latent modes, VIV is confounded by nominal dimensionality and by dead (near-zero-variance) channels: a near rank-deficient latent whose

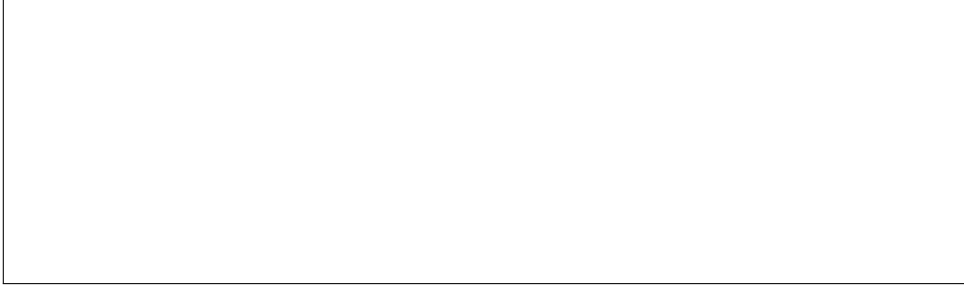


Figure 11: **TODO: Ray-addressed content reuse (qualitative).** On a camera trajectory that leaves a region and later returns, content generated the first time a ray is seen can reappear when the same ray is revisited. A qualitative illustration only, not a quantitative metric.

spectrum is dominated by near-zero eigenvalues has its $\mathbb{E}[\sqrt{\lambda}]$ dragged down and thus *wins* on VIV without being genuinely smooth.

- ρ : within-class velocity variance as a fraction of total; dimension-free and therefore comparable across encoders regardless of channel count. It is the natural companion to VIV once the scale confound must be removed.
- **eVIV**, ρ_e , ϕ , r_e : VIV restricted to each encoder’s own *effective* channel subspace — the top- r_e marginal-covariance directions that cover 99% of channel variance — which removes dead-channel dilution. ρ_e is its dimension-free companion and $\phi = \text{eVIV} / (\frac{\pi}{2} \sqrt{\rho_e})$ is a spectrum-shape proxy. Since r_e differs across encoders, eVIV magnitudes are only comparable at equal r_e ; ρ_e and ϕ carry the ranking.
- **gVIV** (matched-dimension / “gold” VIV): VIV after projecting every latent onto a *common* fixed $16 \times 16 \times 16$ subspace (top-16 channel PCA followed by spatial pooling), giving a directly cross-encoder-comparable number that neutralizes the dimensionality confound.

Spectrum / conditioning.

- **SEC_q** (spectral energy concentration): fraction of latent energy carried by the high spatial frequencies (the complement of the lowest- q band); we report SEC_{.25} and SEC_{.50}. Lower means spatially smoother.
- κ : condition number of the latent channel covariance (ratio of largest to smallest eigenvalue); large κ indicates a near rank-deficient, numerically hard-to-train latent.
- **effrank** (effective rank): the participation ratio / spectral-entropy count of “live” channels, i.e. how many of the nominal channels actually carry variance.

Structure (cosine metrics, per-sample centered, $\text{rm_pc}=0$, to remove anisotropy).

- **LNC@ k** (latent neighbor consistency): semantic separability from k -NN label agreement (masked to ImageNet-S foreground).
- **LDS / CDS** (latent / cross distance structure) and **SRSS** (spatial relative similarity score): spatial neighborhood structure, using image-pixel distances (near $\leq 24\text{px}$, far $\geq 96\text{px}$) so scores are comparable across latent grids; SRSS additionally uses a foreground mask.

A.2 COMPLETE PER-ENCODER DIAGNOSTICS

Table 11 reports the diffusability and spectrum diagnostics. Table 12 reports the centered structure diagnostics, while Table 13 shows the raw-to-centered change used to diagnose anisotropy. The main-paper structure columns intentionally use the raw (uncentered) scores, because those are the scores seen by the representation before removing dominant global components.

Table 11: **Diffusability and spectrum diagnostics** (single-view ImageNet, res. 256). Lower is better for $VIV/\rho/eVIV/\rho_e/\phi/gVIV/SEC/\kappa$; higher effrank is better. r_e is the number of effective channels at the 99% energy threshold. Per ranked column, the **best** / **second-best** across latents are shaded.

Latent	$C \times H \times W$	VIV	ρ	eVIV	ρ_e	ϕ	r_e	gVIV	SEC. ₂₅	SEC. ₅₀	effrank	κ
sd-vae	$4 \times 32 \times 32$	5.027	0.888	5.174	0.884	3.503	4	2.100	0.403	0.208	3.0	3.7
wan2.1	$16 \times 32 \times 32$	1.773	0.847	2.472	0.872	1.685	13	0.685	0.093	0.034	5.3	127
dinov2	$1024 \times 18 \times 18$	0.868	0.803	0.929	0.793	0.664	256	0.459	0.364	0.160	239.7	$5.1e13$
raev2	$1024 \times 16 \times 16$	0.768	0.711	0.703	0.746	0.518	256	0.479	0.296	0.132	289.3	$1.8e4$
da3-giant	$3072 \times 18 \times 18$	0.553	0.769	0.649	0.869	0.444	256	0.475	0.044	0.022	11.5	$6.6e16$
ngd-giant	$128 \times 18 \times 18$	0.585	0.674	0.691	0.770	0.501	116	0.530	0.279	0.114	51.3	227

Table 12: **Structure diagnostics** (single-view ImageNet, res. 256; centered cosine variant). Higher is better. Per column, the **best** / **second-best** across latents are shaded.

Latent	LNC@5	LNC@10	LDS	CDS	SRSS
sd-vae	0.047	0.044	0.328	0.0035	0.435
wan2.1	0.084	0.069	0.557	0.0057	0.721
dinov2	0.931	0.901	0.475	0.0043	0.662
raev2	0.930	0.901	0.333	0.0044	0.434
da3-giant	0.063	0.050	0.534	0.0044	0.660
ngd-giant	0.506	0.417	0.586	0.0058	0.719

A.3 WHY THE RECORD-LOW VIV OF RAW GEOMETRY FEATURES IS DECEPTIVE

By definition $VIV = \frac{\pi}{2} \mathbb{E}[\sqrt{\lambda}]$, so a latent whose velocity spectrum is dominated by many near-zero eigenvalues has its mean $\sqrt{\lambda}$ pulled down and posts an artificially low VIV. Raw da3-giant is exactly this case: of its 3072 channels only effective rank ≈ 11 (about 0.4%) carry variance, with a condition number $\kappa \approx 6.6 \times 10^{16}$ (Table 11) — a handful of real directions plus a very long near-zero tail. Three controls confirm that its lowest-in-both-regimes VIV/SEC reflects this spectral degeneracy, not genuine transport-smoothness.

(i) Matched dimension. Projecting every latent onto a common $16 \times 16 \times 16$ PCA subspace (gVIV) removes the dimensionality confound. da3-giant’s large raw-VIV lead over the semantic RAE (dinov2, 0.55 vs. 0.87, a 0.32 gap) collapses to within 0.02 and in fact *reverses*: gVIV 0.475 for da3-giant versus 0.459 for dinov2 (Table 11). The apparent smoothness was bought entirely with extra nominal channels padded by dead modes; at equal dimension raw geometry is no longer the smoothest latent.

(ii) Dimension-free ρ . The within-class variance fraction ρ needs no dimensionality correction. Raw da3-giant is only mid-table ($\rho = 0.77$ single-view), *worse* than our compact ngd-giant ($\rho = 0.67$, best in column). So even before matching dimension, the dimension-free measure already denies raw geometry the “smoothest” crown that raw VIV appears to award it.

(iii) Anisotropy. A spectrum ruled by a few giant directions also shows up in the structure metrics: on *raw* features da3-giant scores near-zero (LDS 0.05, SRSS 0.07), but after removing a single dominant principal component the same scores jump an order of magnitude (LDS $0.05 \rightarrow 0.53$, SRSS $0.07 \rightarrow 0.66$). This raw-to-centered jump is the tell-tale signature of a latent dominated by one or a few enormous anisotropic modes — consistent with effrank ≈ 11 — rather than a smooth, high-rank spectrum.

eVIV corroboration. Restricting VIV to each encoder’s own effective subspace tells the same story: after dead channels are excluded, da3-giant’s eVIV increases relative to its raw VIV ($0.553 \rightarrow 0.649$), and its effective-subspace variance fraction is high ($\rho_e = 0.869$), so the raw-VIV ranking no longer supports the interpretation that raw geometry is uniquely smooth. Taken together, all four de-confounding views (gVIV, ρ , anisotropy, eVIV) agree: raw geometry features

Table 13: **Effect of per-sample centering on structure metrics.** Each entry is raw / centered. Large jumps after centering indicate that a few dominant anisotropic directions obscure the local structure.

Latent	LNC@5	LDS	SRSS
sd-vae	0.051 / 0.047	0.262 / 0.328	0.324 / 0.435
wan2.1	0.084 / 0.084	0.234 / 0.557	0.313 / 0.721
dinov2	0.931 / 0.931	0.365 / 0.475	0.519 / 0.662
raev2	0.932 / 0.930	0.171 / 0.333	0.220 / 0.434
da3-giant	0.054 / 0.063	0.054 / 0.534	0.070 / 0.660
ngd-giant	0.498 / 0.506	0.444 / 0.586	0.553 / 0.719

are *not* genuinely the most diffusable latent; their raw-VIV advantage is a dimensionality / dead-channel artifact. Compressing them into the compact `ngd-giant` latent removes the degenerate tail, dropping κ by ~ 14 orders of magnitude and restoring a healthier effective rank while keeping VIV low.

B ADDITIONAL QUALITATIVE COMPARISONS

Figures 12–21 expand the compact panels of Figures 5 and 6 to all nine views of a single scene, so the full camera trajectory is visible for every latent. All rows come from the same matched 64-scene / 9-view pools used by Tables 4–6: one reference view, identical metric ray-space camera conditioning, 50 Euler steps, and $\text{CFG} = 2$. Because the scene manifest is shared, a given column is the same target viewpoint for every method. Rows are the seven latents plus the two external systems `GLD` and `Gen3R`. The PSNR annotation under each row label is that row’s PSNR on the eight generated views of the displayed scene; it ties the qualitative behaviour to the aggregate numbers and is not a scene-averaged score. Note that the reference column is itself a generated slot under our clean-token formulation (Section 4.3) rather than a copy of the input.

We state how the scenes were chosen, because both conditions we impose matter. First, camera motion has to be substantial: we measure it as the largest distance between two camera centres divided by the median scene distance, both taken from one Pi3 forward pass so the ratio is scale-free, and keep only clips above the 60th percentile of the pool. Second, *both* NGD rows have to be ahead of all seven other rows on PSNR, not just the stronger of the two. Neither condition alone suffices. Ranked on margin alone the selection drifts to near-static clips, where all nine views are nearly the same frame, every method scores well and nothing is really being synthesized; ranked on motion alone it drifts to clips where only NGD-64 stays ahead. Under both conditions we show six RealEstate10K scenes, spanning 0.40 to 0.60 scene distances of camera motion with margins of +0.39 to +1.38 dB for the weaker NGD row.

DL3DV gets four scenes rather than six, and the reason is itself informative: NGD ($C=128$) is the weaker of our two latents on DL3DV (Table 4), and on several of the widest-motion DL3DV clips it falls behind SD-VAE, so those clips fail the second condition. Rather than relax either condition we report fewer DL3DV scenes; their motion spans 0.28 to 0.61 scene distances.

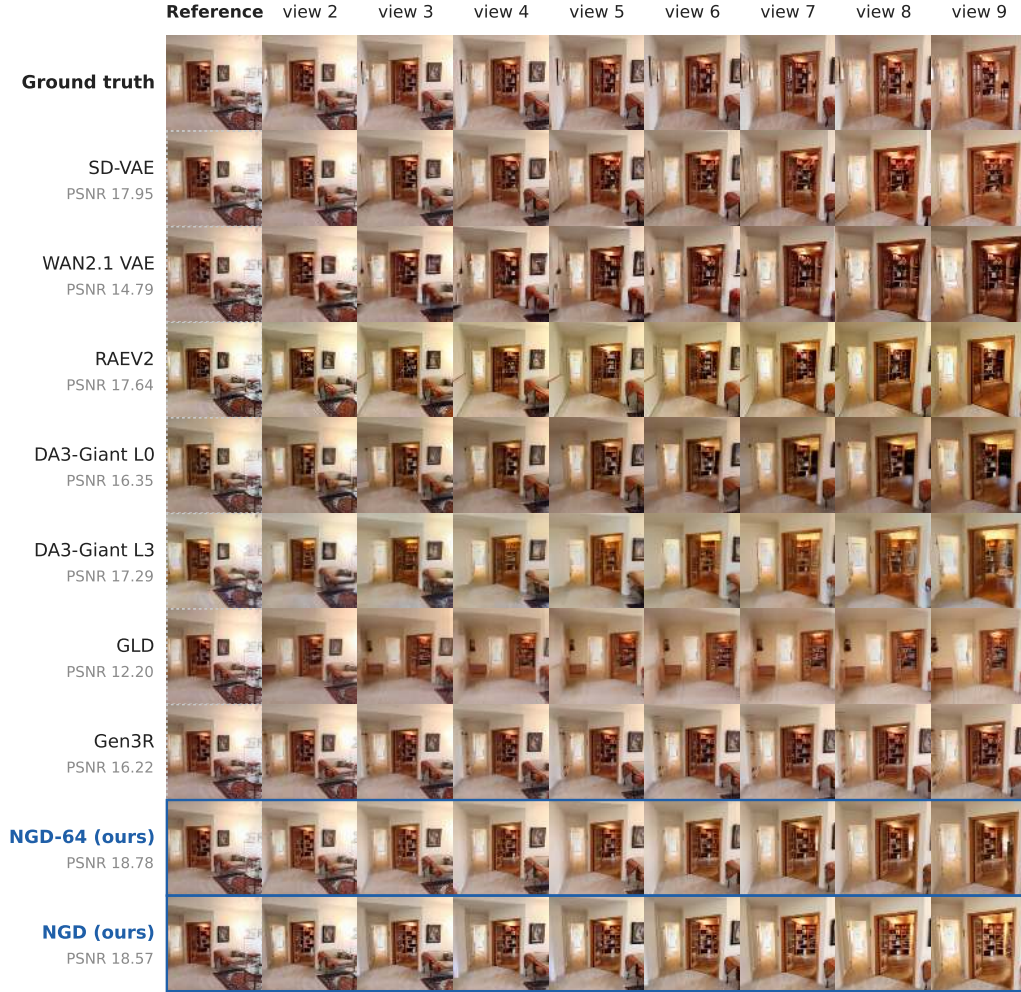


Figure 12: **Per-scene latent comparison on RealEstate10K (scene 024).** All nine views; the widest camera motion of the six (0.60 scene distances, 89th percentile). A hallway turn, which is the motion that most exposes pose drift: GLD loses the doorway and the framed pictures on the left wall, DA3-Giant L0 keeps them but shifts them across the frame, and both NGD rows (18.8 and 18.6 dB) stay ahead of SD-VAE at 18.0 dB.

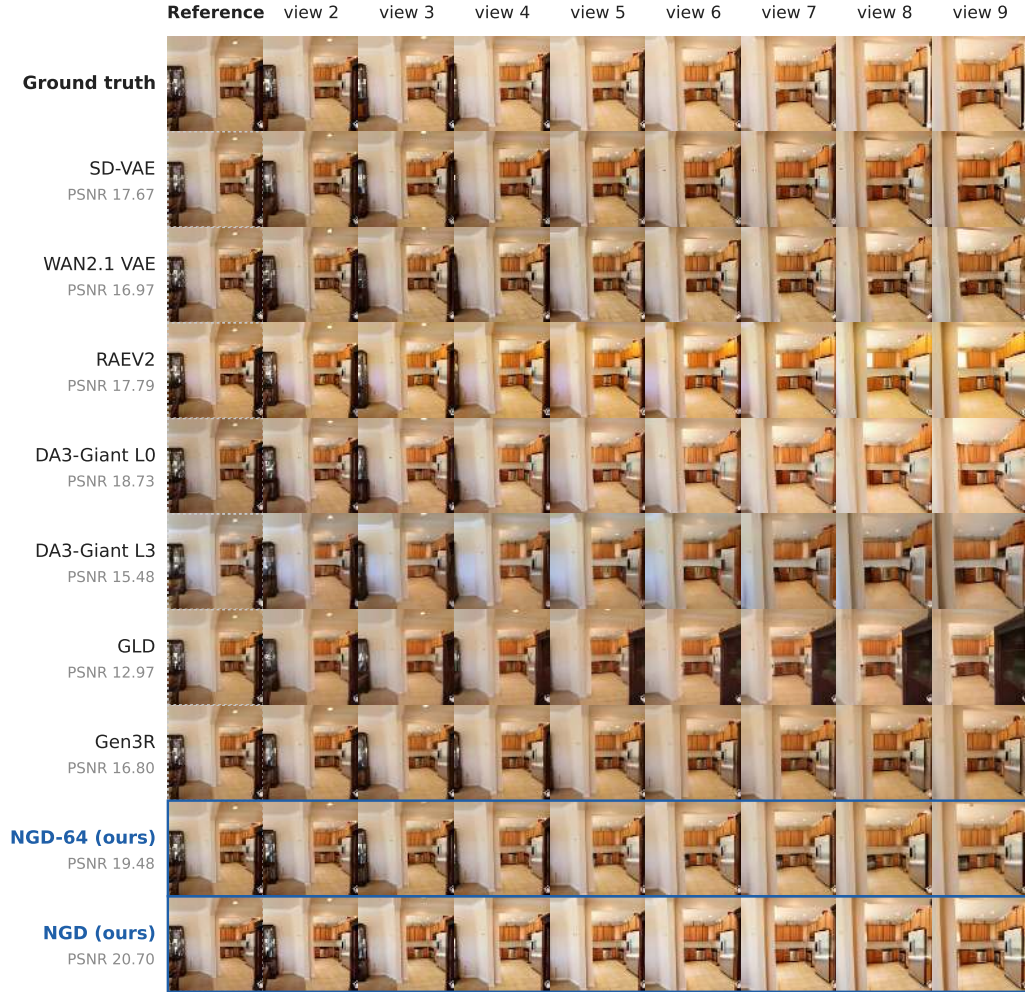


Figure 13: **Per-scene latent comparison on RealEstate10K (scene 002)**. Camera motion 0.57 scene distances. Translation down a kitchen, where the cabinet faces and the counter edge are the structures that reveal drift. NGD ($C=128$) leads NGD-64 here by 1.2dB, the largest of the three scenes of the ten where the wider latent comes out ahead; the strongest baseline is DA3-Giant L0 rather than SD-VAE.

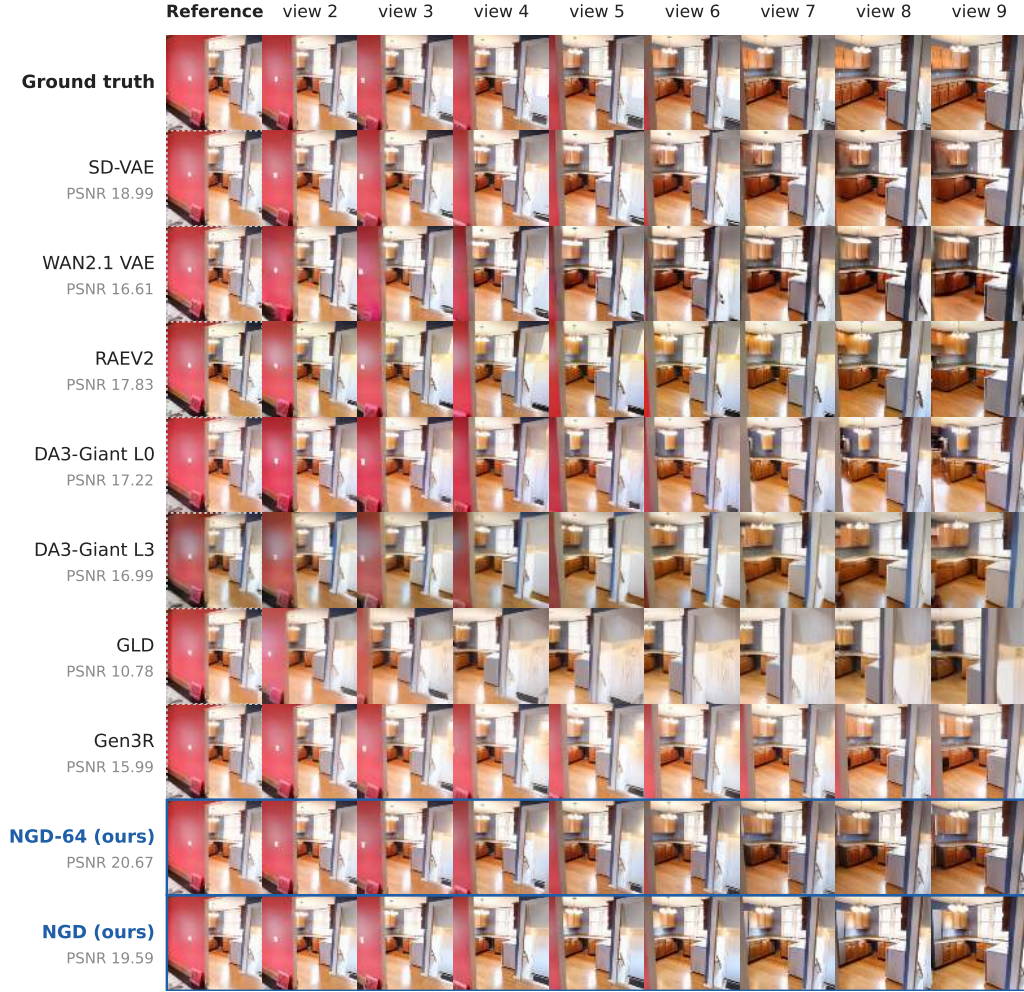


Figure 14: **Per-scene latent comparison on RealEstate10K (scene 052)**. Camera motion 0.49 scene distances. The red accent wall is a strong colour anchor and the camera turns off it into a kitchen: GLD dissolves the wall into a white haze by view 4 (10.8 dB, the weakest row of any scene shown here), while both NGD rows carry the wall and the pendant light through the turn.

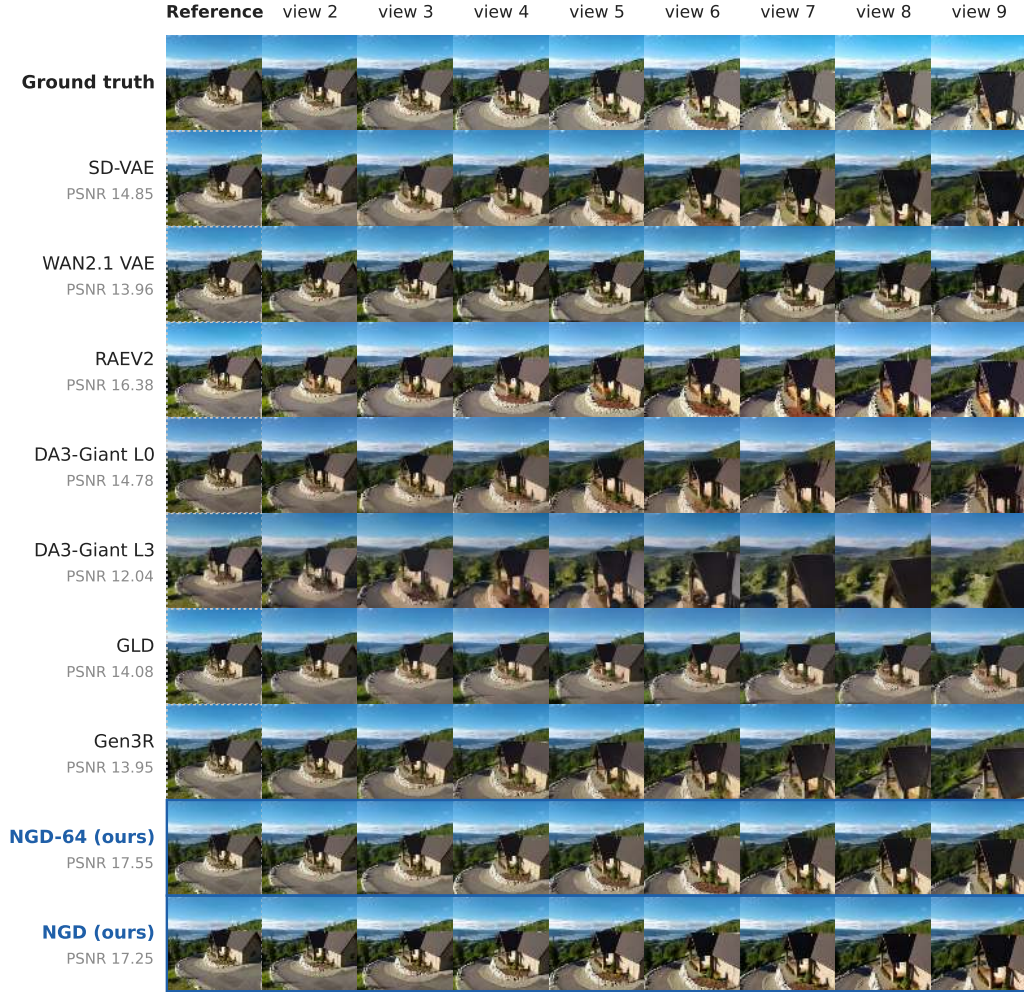


Figure 15: **Per-scene latent comparison on RealEstate10K (scene 044)**. Camera motion 0.47 scene distances. The only exterior scene in these panels, and an orbit rather than a dolly: the roof line has to stay straight against the sky while the curved driveway sweeps past. DA3-Giant L3 blurs the whole scene (12.0 dB) and WAN2.1 and Gen3R lose the roof edge, whereas both NGD rows hold it. RAEV2 is the strongest baseline here, which fits its role as the semantic latent.

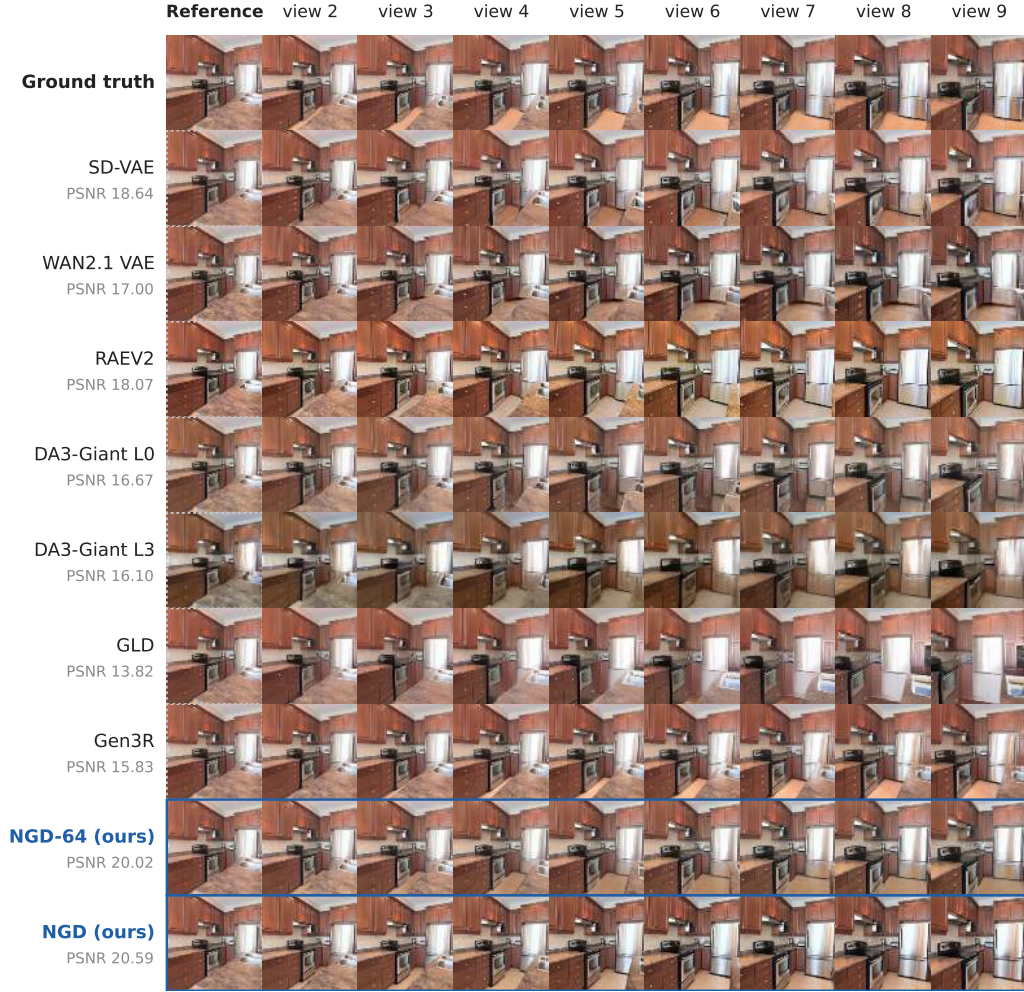


Figure 16: **Per-scene latent comparison on RealEstate10K (scene 020)**. Camera motion 0.41 scene distances, and the largest margin of the six (+1.4 dB for the weaker NGD row). Dark wood cabinets next to a specular stainless kettle put a wide dynamic range in one frame: GLD flattens the cabinets (13.8 dB) and Gen3R loses the kettle highlight, while both NGD rows stay above 20 dB.

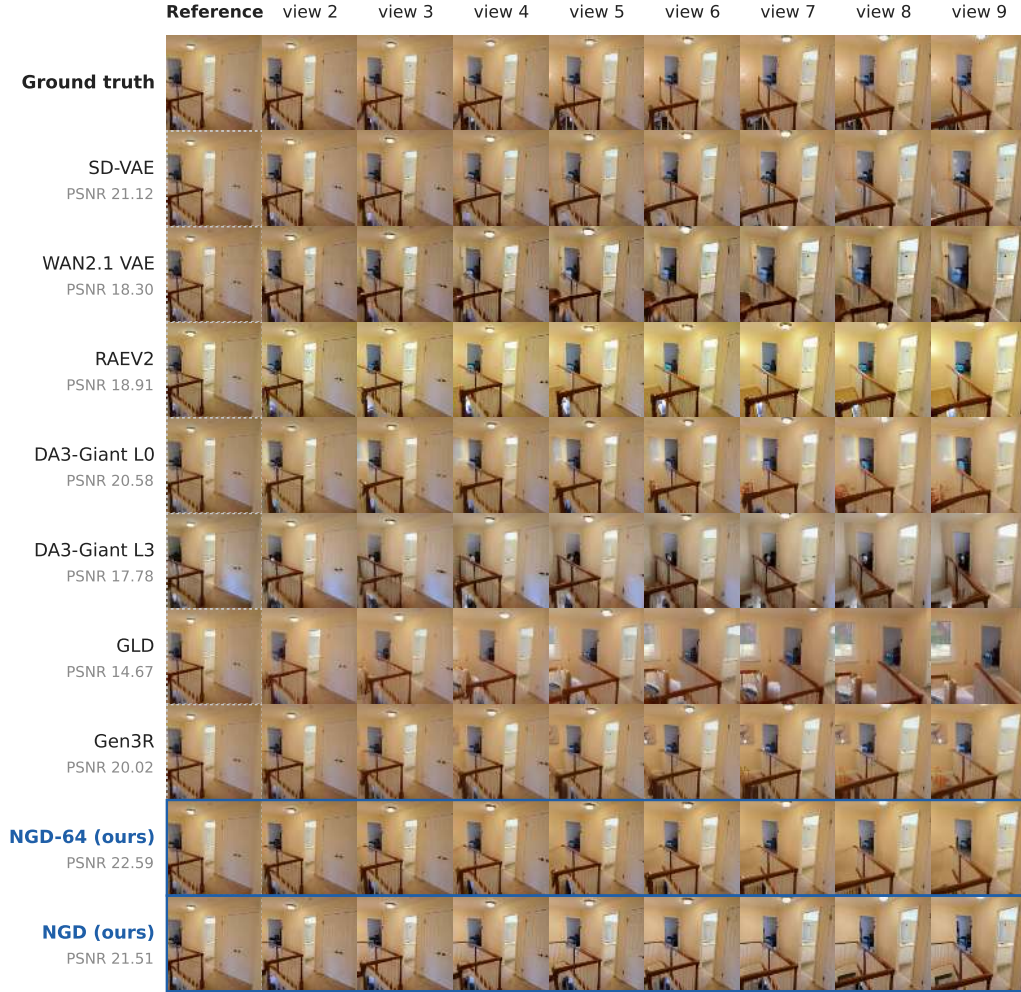


Figure 17: **Per-scene latent comparison on RealEstate10K (scene 041).** Camera motion 0.40 scene distances, and the narrowest margin of the ten (+0.39 dB). An easy scene in absolute terms — SD-VAE, DA3-Giant L0 and Gen3R are all above 20 dB — included to show that the selection is not driven only by baseline failures. The wooden banister is the diagnostic structure: GLD loses it entirely (14.7 dB), and NGD-64 leads at 22.6 dB.

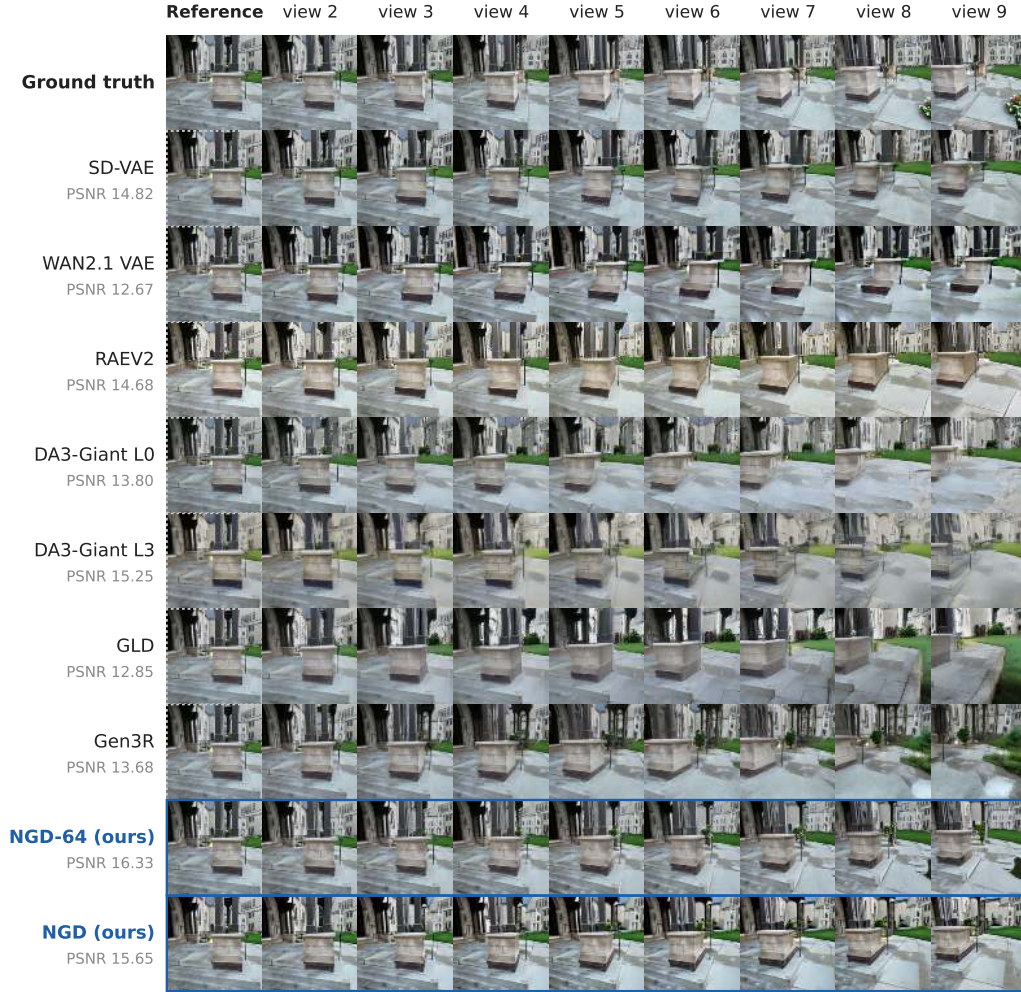


Figure 18: **Per-scene latent comparison on held-out DL3DV (scene 019)**. All nine views; the widest camera motion in the DL3DV panels (0.61 scene distances, 92nd percentile). An outdoor scene, out of domain for the indoor-heavy training mix, and the hardest of the four in absolute terms: the stone facade and the lawn boundary drift in GLD (12.8 dB) and Gen3R, while both NGD rows stay ahead of a comparatively strong DA3-Giant L3.

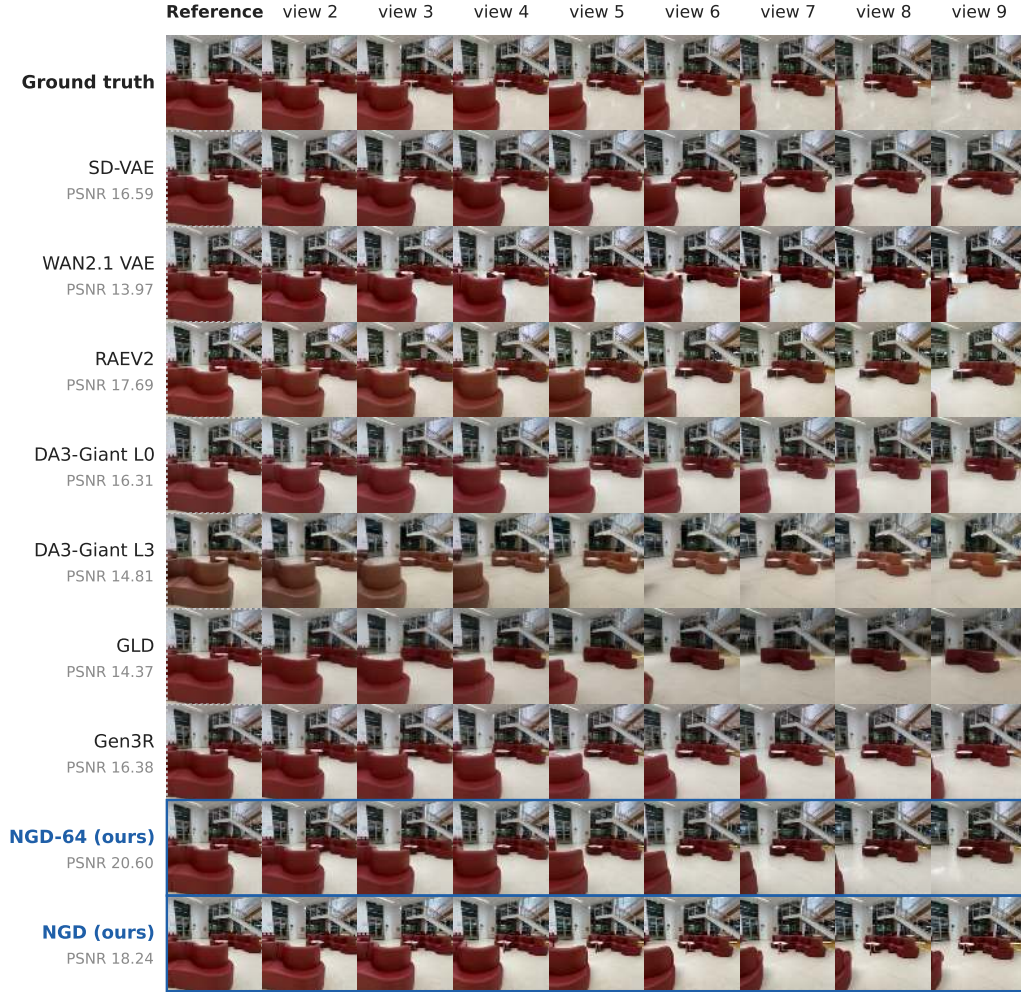


Figure 19: **Per-scene latent comparison on held-out DL3DV (scene 038).** Camera motion 0.35 scene distances, and the largest single-row margin of any scene shown (20.6 dB for NGD-64 against 17.7 for the best baseline). The red sofa against a glazed wall separates the methods cleanly: WAN2.1 and DA3-Giant L3 smear the sofa into the floor, GLD and Gen3R bend the window frames, and NGD-64 keeps both.

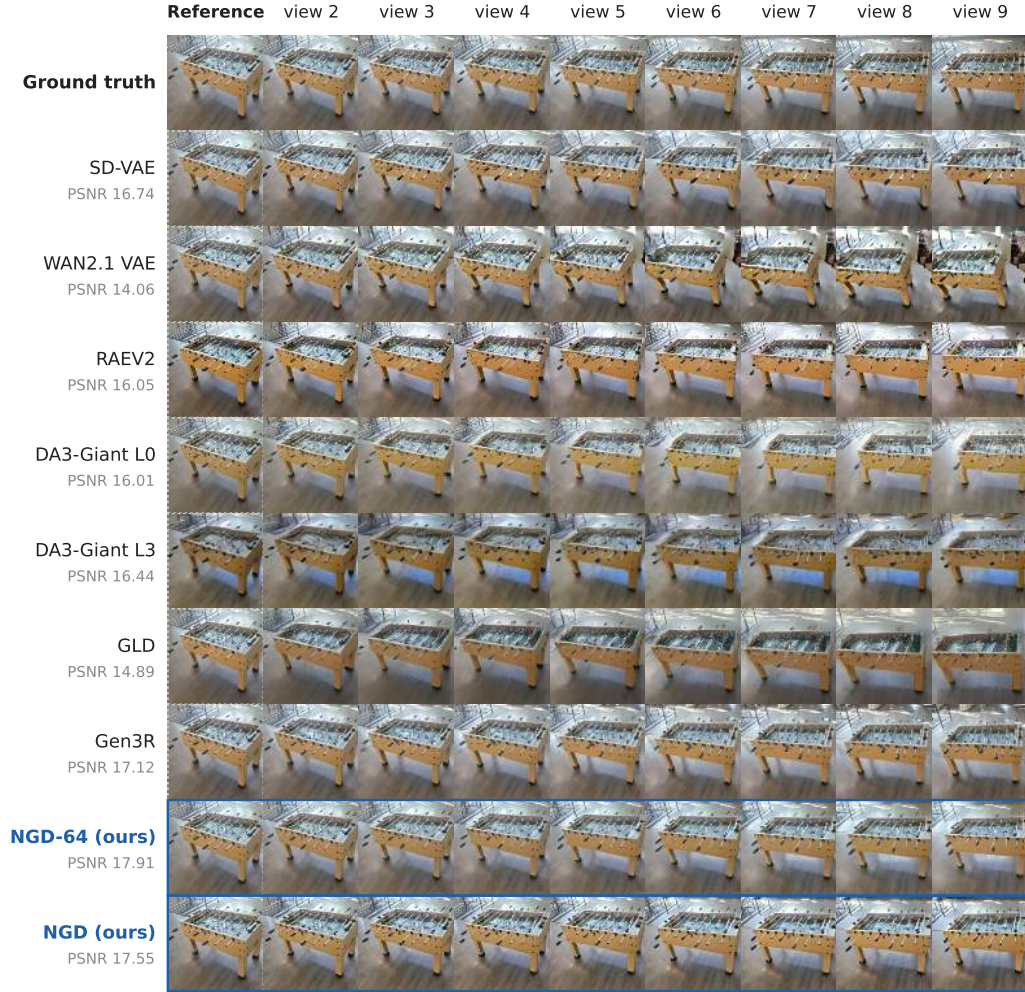


Figure 20: **Per-scene latent comparison on held-out DL3DV (scene 015).** Camera motion 0.31 scene distances. This is the same scene as the DL3DV geometry panel (Figure 24), so appearance and decoded geometry can be read against each other: the thin rods of the table are what both the RGB rows and the depth rows have to preserve. Gen3R is the strongest baseline here on appearance (17.1 dB) despite its weak trajectory.

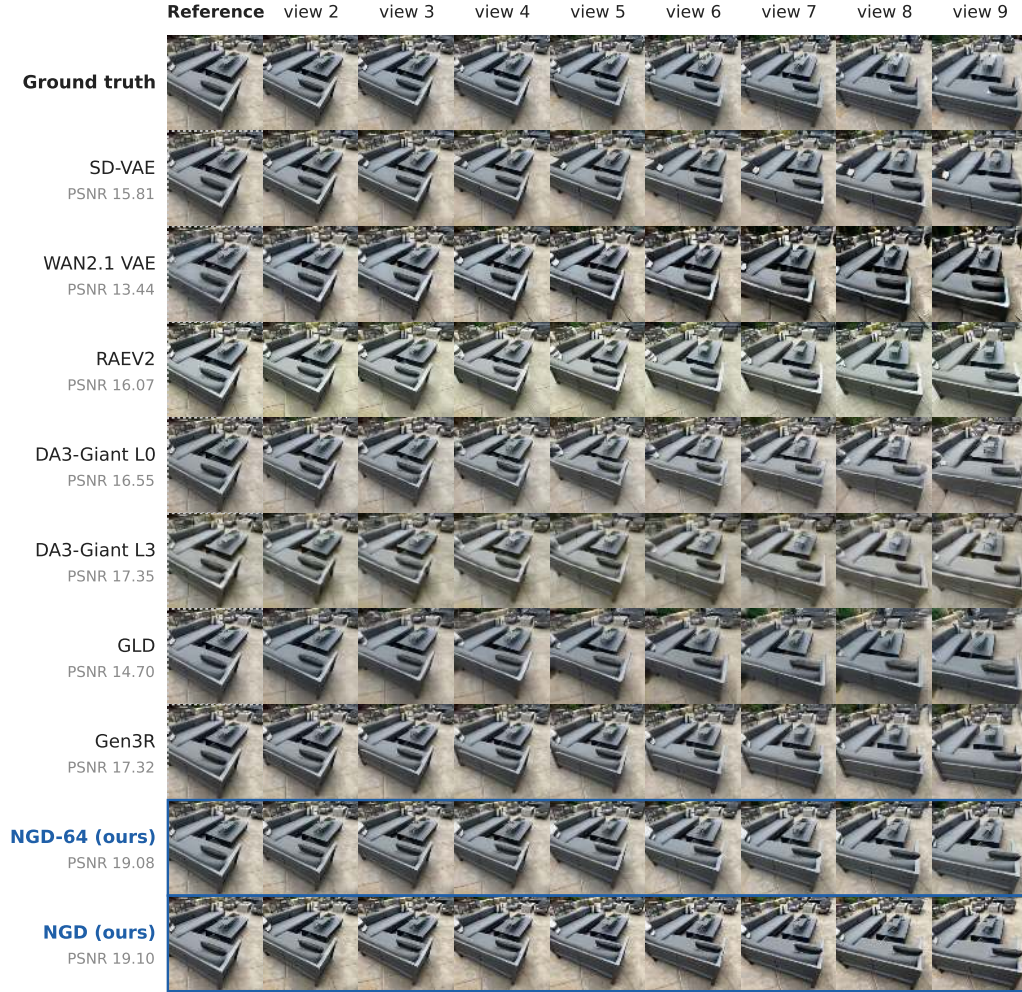


Figure 21: **Per-scene latent comparison on held-out DL3DV (scene 032).** Camera motion 0.28 scene distances, and the largest margin of the ten (+1.7 dB for the weaker NGD row). A row of nested grey seating units is almost pure repeated structure, which is exactly what invites a latent to duplicate or drop an instance: WAN2.1 (13.4 dB) and GLD lose the count, and the two NGD rows agree to within 0.02 dB, the closest the two latent widths come on any scene shown.

Figures 22–25 are the per-scene versions of Figure 8. They drop the oracle row and show five evenly spaced views rather than seven, which buys roughly double the cell size: unlike the appearance panels, what has to be legible here is fine depth structure, not the length of the trajectory. Reading the three outputs together separates two failure modes the aggregate numbers of Table 9 conflate: a latent can hold its camera path while its depth loses scale (DA3-Giant L0 at wide baselines), or keep plausible depth while the path drifts (Gen3R once the camera turns).

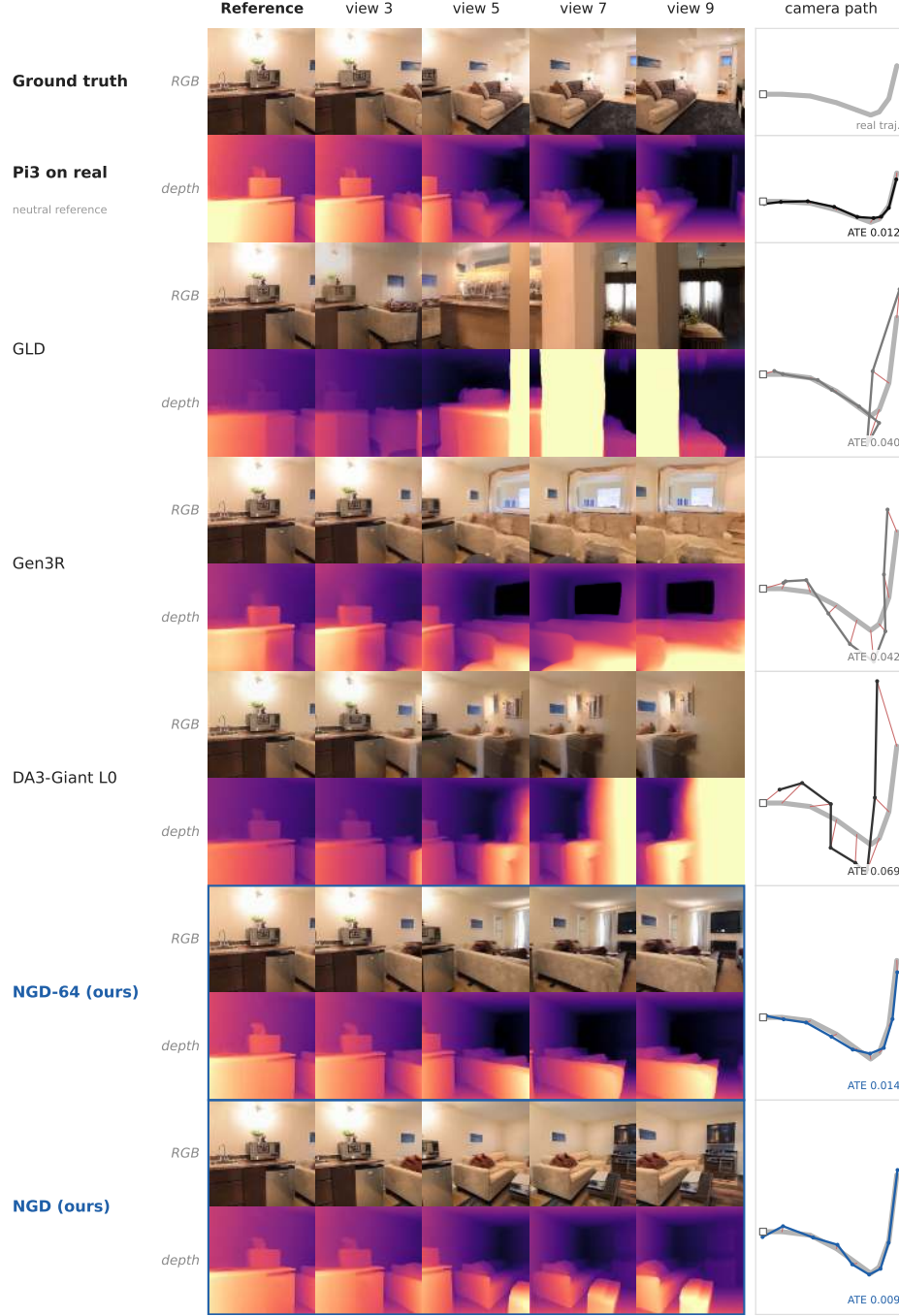


Figure 22: **Jointly decoded RGB, geometry and camera path on RealEstate10K (scene 035).** All geometry latents on the scene of Figure 8. Depth shares one colour map after median scale alignment to the Pi3 row, and camera paths are Sim(3)-aligned to the real trajectory with the cross-track axis magnified. GLD and Gen3R break down in the second half of the trajectory in both appearance and depth, and DA3-Giant L0 additionally leaves the real camera path; both NGD rows stay on the trajectory and keep the sofa and cabinet geometry.

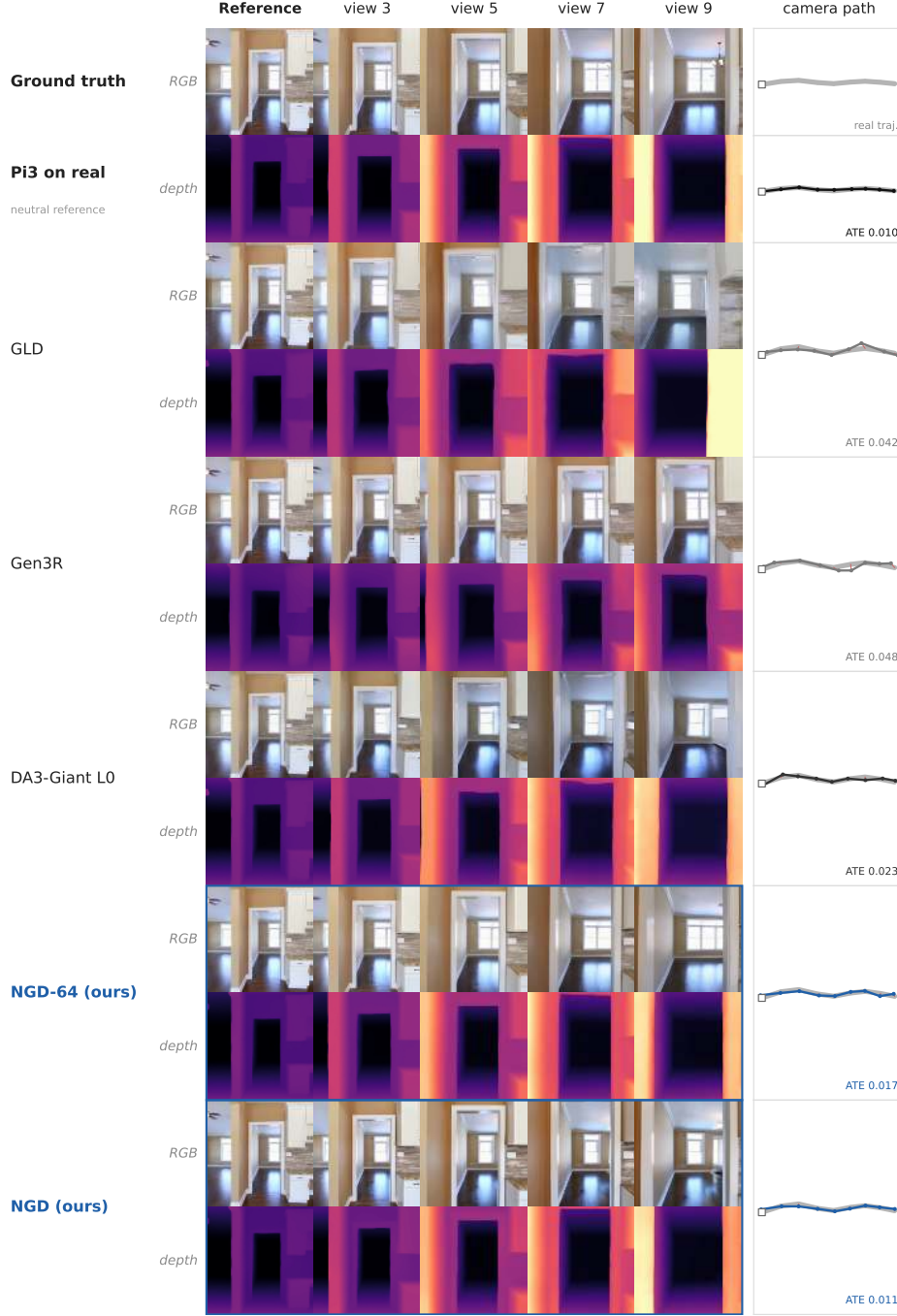


Figure 23: **Jointly decoded RGB, geometry and camera path on RealEstate10K (scene 010).** Same protocol as Figure 22, on the widest camera motion in the RealEstate10K pool (0.81 scene distances, 95th percentile). The doorway is a far-depth opening framed by near walls, so the decoded depth has to hold a large discontinuity while the camera translates through it: GLD’s depth saturates in the last view and Gen3R has the worst trajectory of any row (ATE 0.048), while NGD ($C=128$) at 0.011 is within 0.001 of Pi3 reading the real frames.

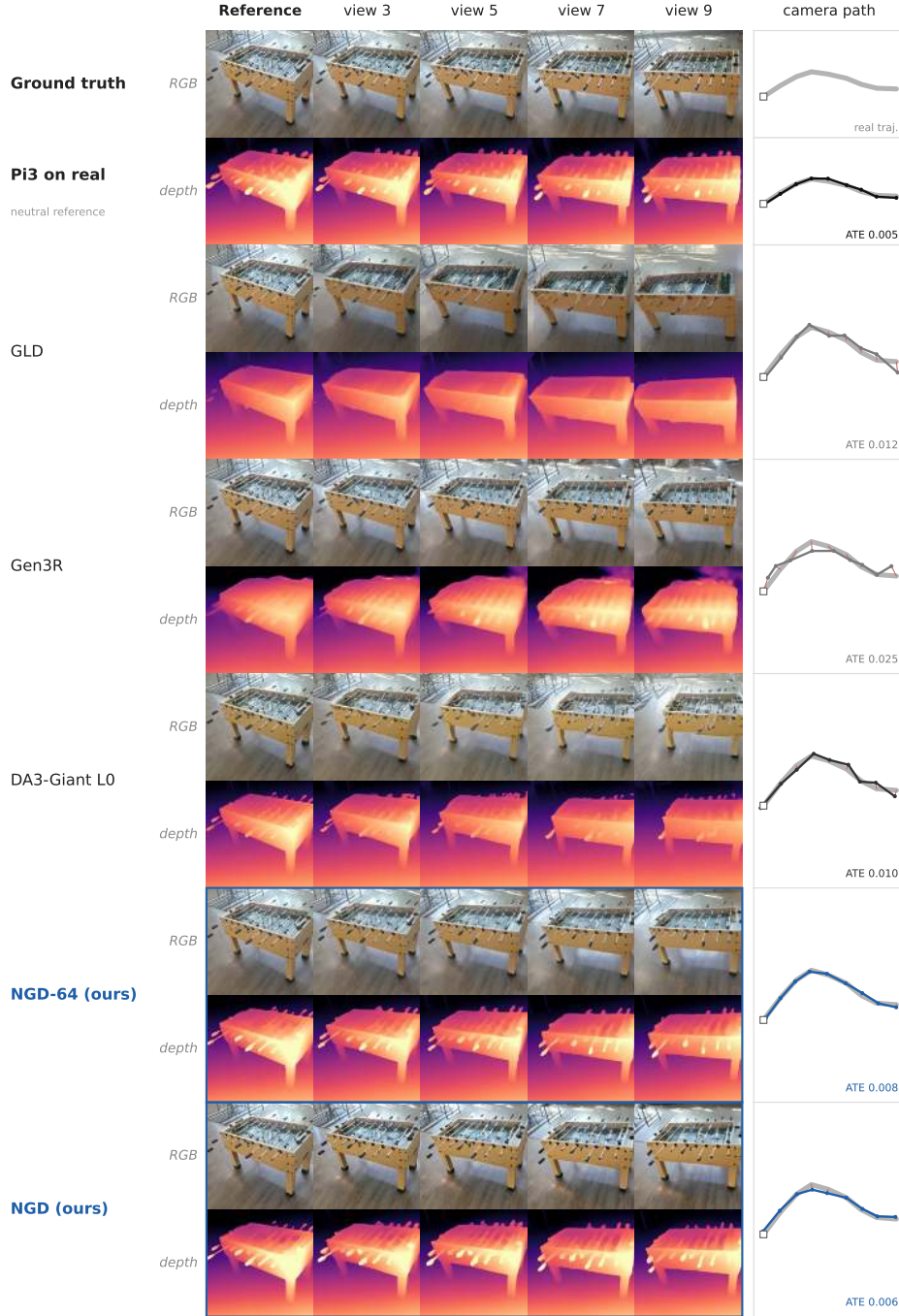


Figure 24: **Jointly decoded RGB, geometry and camera path on held-out DL3DV (scene 015).** Same protocol as Figure 22. The table’s thin rods are the diagnostic structure here: the Pi3 reference row resolves them individually, both NGD rows keep them visible in the decoded depth, and GLD, Gen3R and DA3-Giant L0 flatten them into the table top.

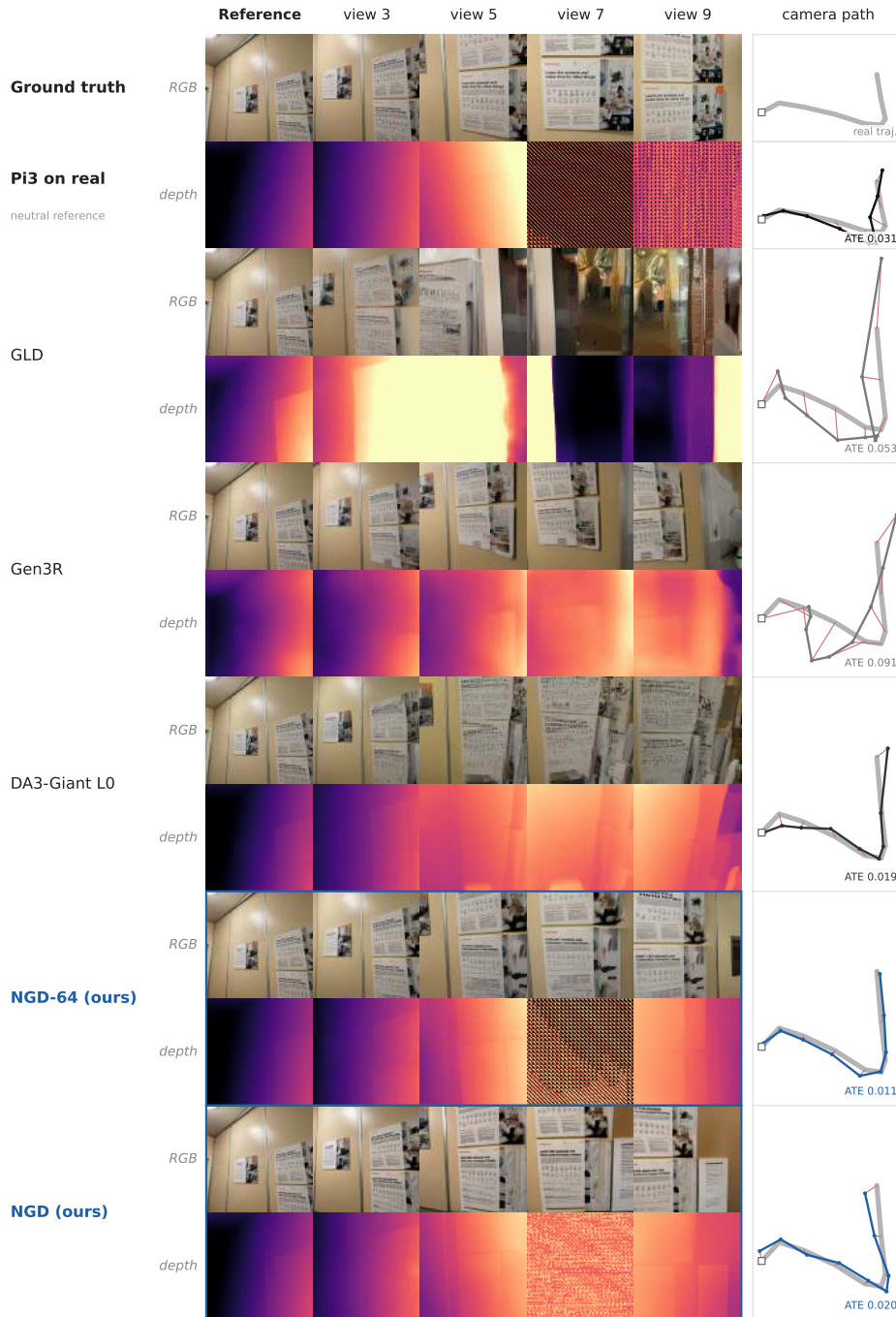


Figure 25: **Jointly decoded RGB, geometry and camera path on held-out DL3DV (scene 039).** Same protocol as Figure 22, on the widest camera motion in the DL3DV pool (0.77 scene distances, 97th percentile), where the trajectory turns through a right angle. This is the cleanest separation of depth quality from pose quality in the pool: Gen3R produces a plausible poster wall and a plausible depth map yet has the worst trajectory of any row (ATE 0.091 against NGD-64’s 0.011), GLD loses both and its depth saturates, and DA3-Giant L0 lands in between (0.019). Sharp turns are where a latent that only interpolates appearance is exposed.

C LONG-ROLLOUT SHOW CASES

Figures 26–37 expand Figure 10: each page shows twelve of the 33 generated views of one scene over three orbits of the point cloud decoded from those same latents. The setting is the one described in Section 4.3: 33 views at 504^2 , one reference view, 50 Euler steps, $\text{CFG} = 2$, EMA weights at DiT step 36k, and a decoder whose geometry path was fine-tuned for 6k steps on cached sampled latents from five domains (DL3DV, MVS-Synth, RealEstate10K train and test, ScanNet++).

The pool is 32 scenes, eight per domain from DL3DV, the RealEstate10K test split, ScanNet++ and MVS-Synth, and the twelve shown were picked by eye from all of them: this is a show case, not a ranking.

The outdoor scenes need one extra step before they can be framed at all, and it is worth stating what it is. On the MVS-Synth street scenes the depth head places sky at $1.4\text{--}4.4\times$ the median scene distance, depending on the clip — far enough to dominate the extent of the cloud, but not beyond the real content, whose own far tail reaches $3\text{--}4\times$. A distance threshold therefore cannot separate the two: cutting far enough to remove sky also removes the building facades, as the street scenes are long and the buildings sit at the end of them. We instead drop points whose colour is sky-like (bright, blue, unsaturated), which on these clips is unambiguous and which flags 0% of the points on every indoor scene, so it is only ever active where it is needed. The same geometry — a long strip with the facade at one end — also makes the cloud too wide to frame, and for the same reason it cannot be trimmed radially. We therefore crop it along its dominant direction instead, keeping the central 0.8 standard deviations of the street axis, which drops the receding tail and leaves the block with the road in front of it. The DL3DV temple yard (Figure 28) has an overcast white sky that the colour rule does not catch, but there the sky does sit in the far tail, so cutting the radial distribution at the 88th percentile suffices — against 97–98 for the indoor scenes, where the cut only removes stragglers.

The renderer is a plain splat rasterizer with a painter’s depth buffer at $2\times$ supersampling; points sitting in a voxel that holds fewer than three points (voxel side 1.2% of the scene radius) are dropped as depth speckle. Orbit angles are per scene because no single angle works for all: a room seen from above is a ceiling, and a room seen from the side is a wall. The street scene is a ground plane plus one building wall, so looking up from below grazes the ground into an underside and flattens the thin wall, while a full top-down loses the facade; the first two views sit above the block and look down at about 25° so the facade stands with the road in front, and the third is a near-top-down street plan. Depth is decoded at 504^2 and unprojected on a stride-4 grid, so each cloud is about half a million points over 33 views.

For reference, on the eleven real-scene panels the geometry decode of the sampled latent agrees with the frozen decode of its re-encoded counterpart to 0.007–0.030 scale-aligned AbsRel with Sim(3) ATE 0.001–0.020. The MVS-Synth panel is the outlier at 0.080 AbsRel and 1.7° relative rotation error, and that is a property of the domain rather than of the scene we picked: averaged over all eight clips of each domain in this pool, the same agreement is 0.020 AbsRel / 0.69° on ScanNet++, 0.025 / 0.15° on RealEstate10K and 0.032 / 0.38° on DL3DV, against 0.074 / 3.13° on MVS-Synth. Note that all of these compare against the fine-tuning target rather than against ground truth, so we quote them only to indicate that the clouds are not drifting away from the decoder’s own reconstruction regime over 33 views.

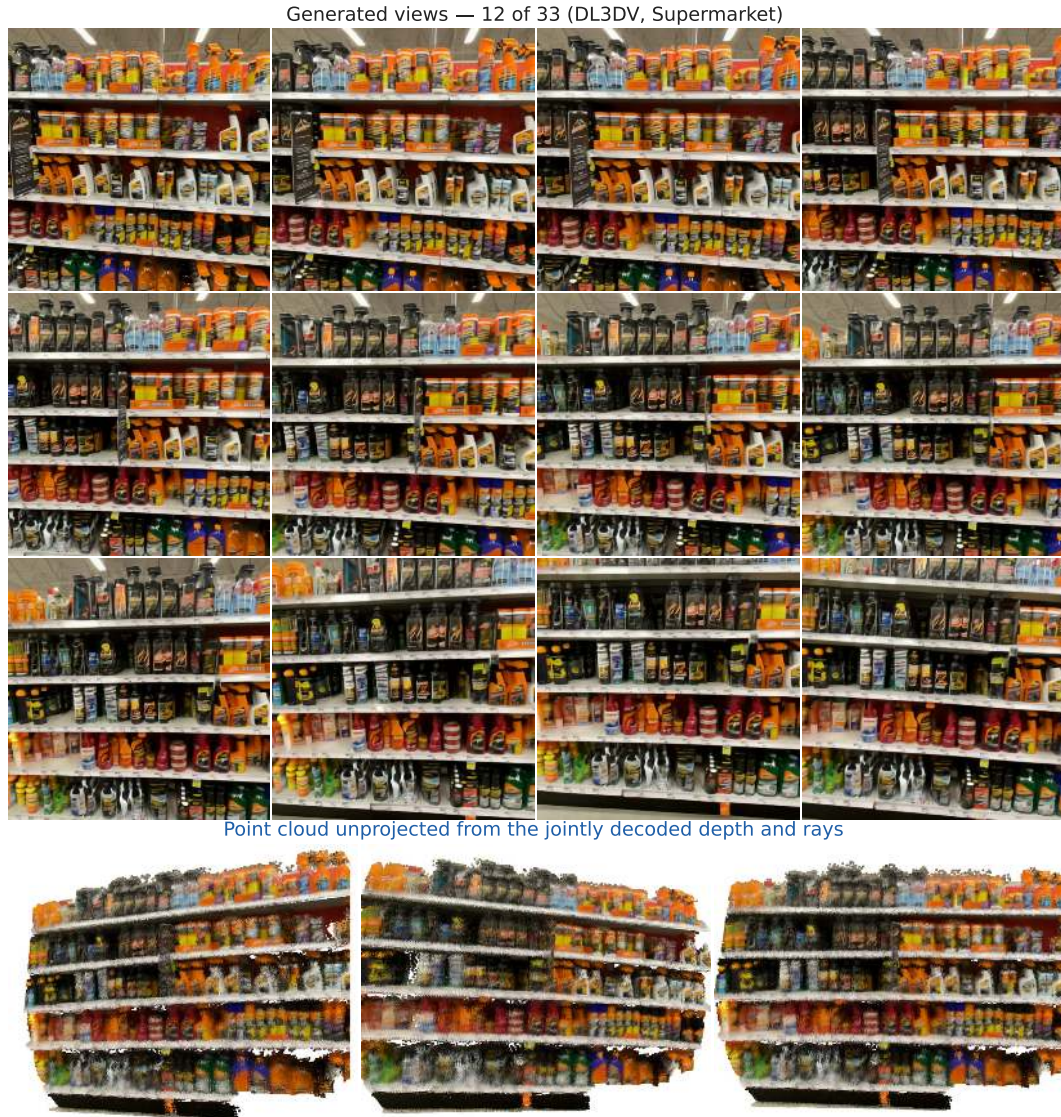


Figure 26: **Long-rollout show case: DL3DV supermarket aisle.** A lateral translation along shelving, which is a useful case because the scene is almost all repeated texture at three distinct depths. In the decoded cloud the shelf faces stay on three separate planes rather than collapsing into one, and the aisle recedes to the right; product labels stay legible in the generated views across the whole rollout.

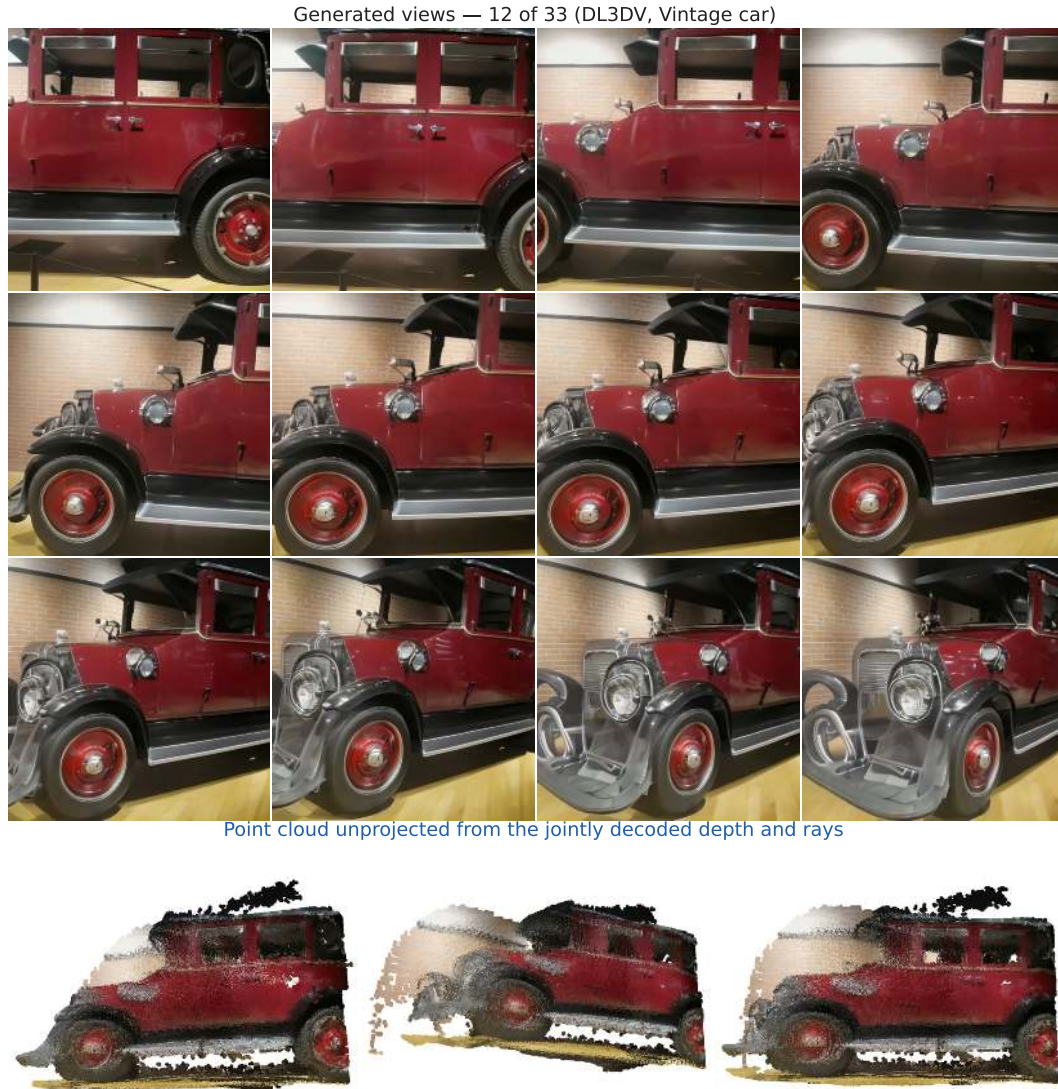


Figure 27: **Long-rollout show case: DL3DV vintage car.** The one object-centric case here: the camera moves from the flank to the front, so the generated views have to keep a rigid object consistent under a large viewpoint change. The cloud holds the body panels, the running board and both wheels as separate surfaces on one ground plane. The dark speckle above the roof is where depth is least certain, at the boundary between the car and the far wall.

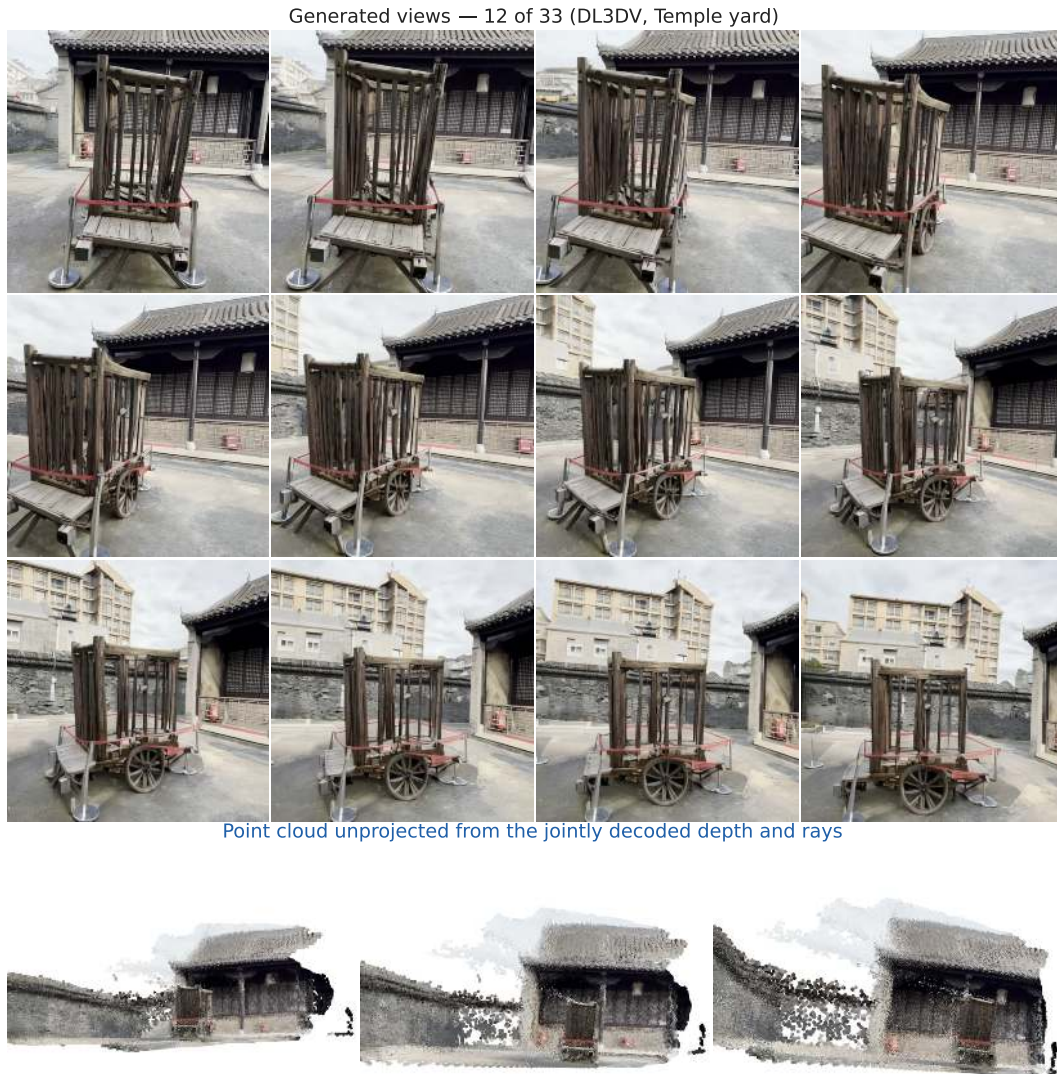


Figure 28: **Long-rollout show case: DL3DV temple yard.** The only real outdoor scene shown; its sky is overcast white rather than blue, so it is trimmed by radial distance (88th percentile) rather than by colour. Without that cut the sky dominates the cloud, as described above. The eaves, the wall behind and the cart in front separate cleanly in depth, which is the structure the sky was hiding.



Figure 29: **Long-rollout show case: DL3DV lounge chairs.** Two armchairs and a low table in a bare room. Furniture against a plain wall is the case where a latent that only interpolates appearance tends to flatten objects into the background; here the chair backs stand off the wall and the table stays a separate volume.

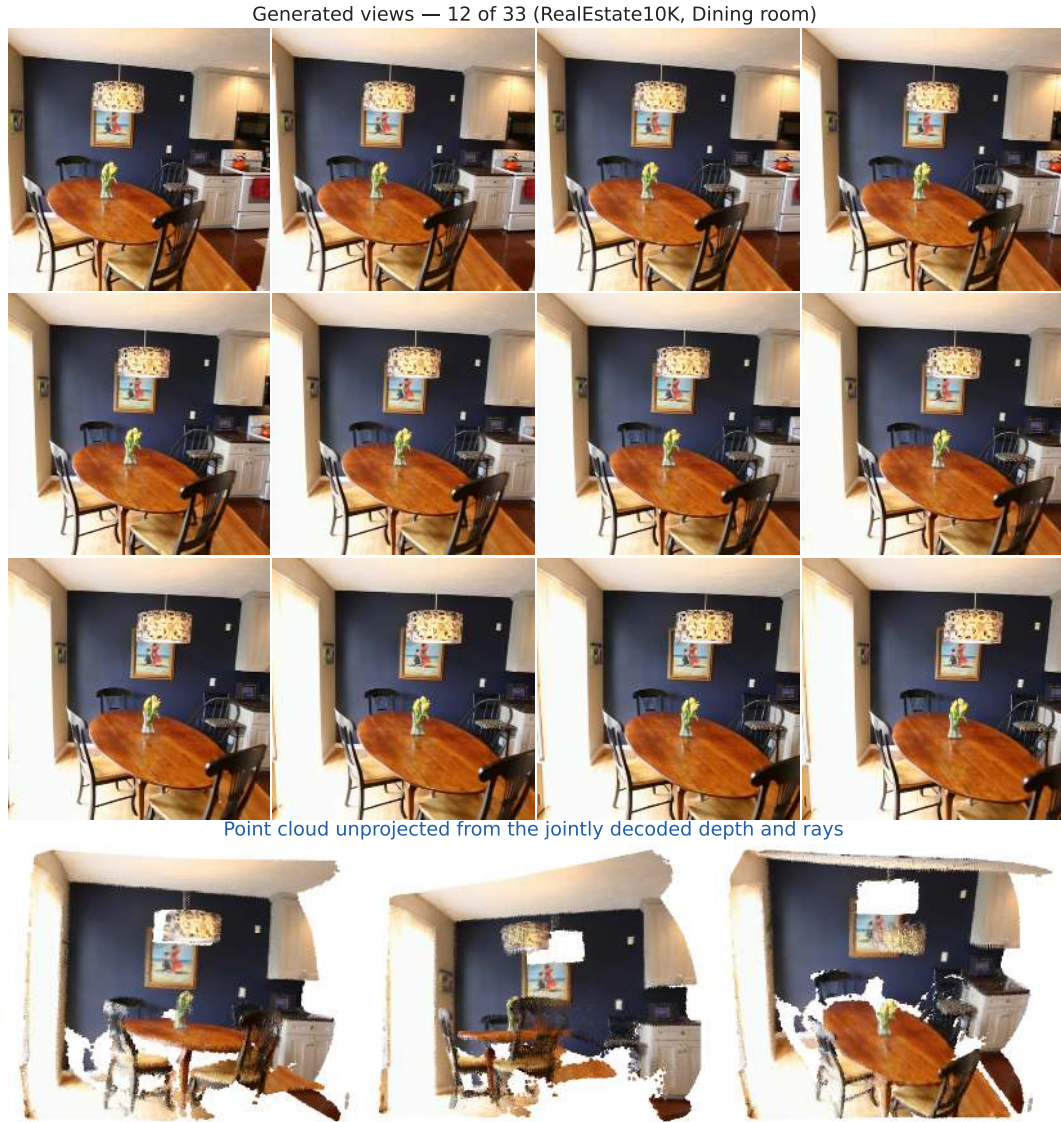


Figure 30: **Long-rollout show case: RealEstate10K dining room.** A dolly past a dining table. Two walls meeting at a corner give an unambiguous check on the decoded geometry, and they meet at a right angle in the cloud; the table, the chair backs, the pendant lamp and the framed picture all survive as separate structures.

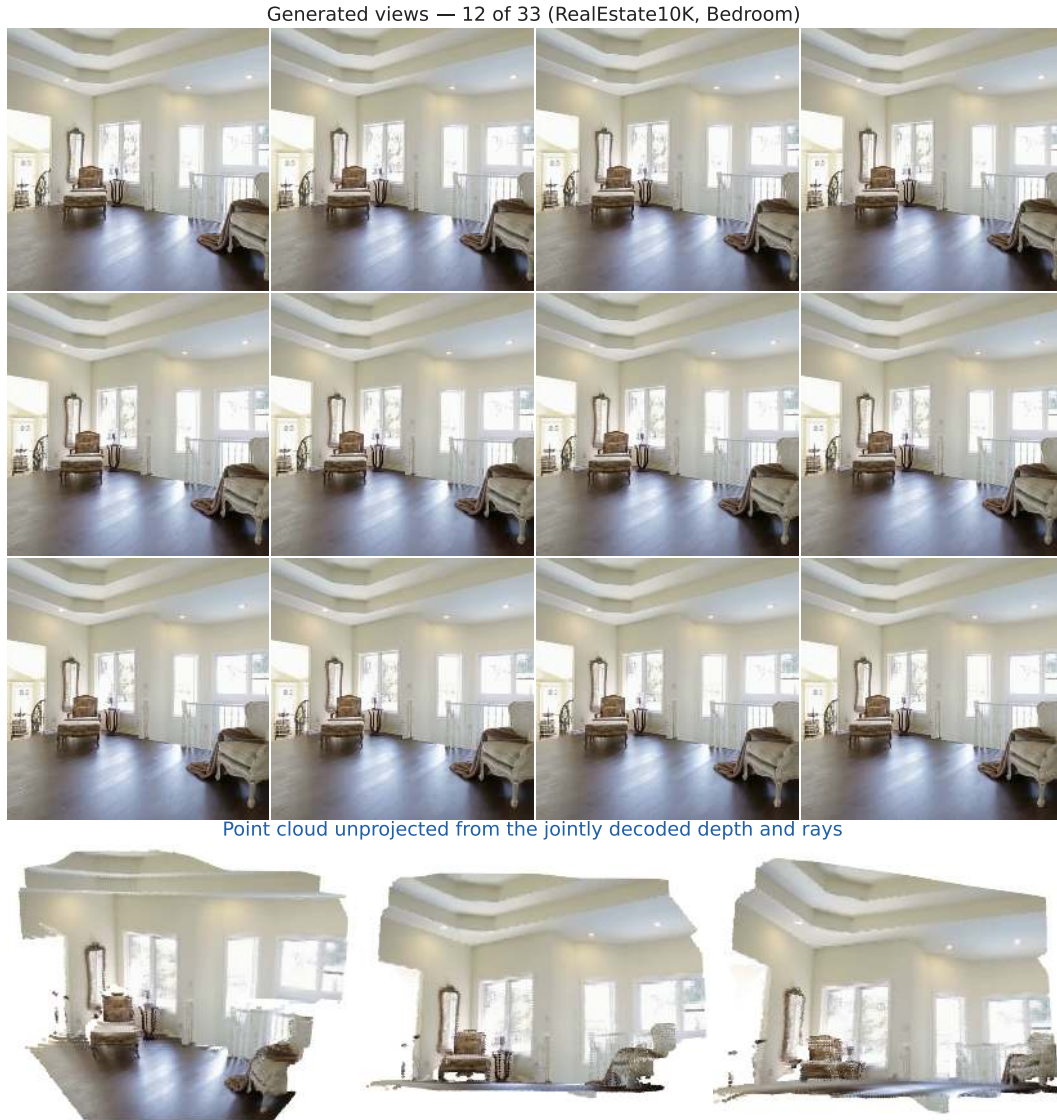


Figure 31: **Long-rollout show case: RealEstate10K bedroom.** The camera barely moves in this clip, so the twelve views look nearly identical — included as the opposite end of the range from the supermarket and car scenes, where the point cloud still has to come out of 33 views whose baselines are small. Bright windows are the hard part: they saturate the RGB decode and, in the cloud, the window wall is the noisiest surface.

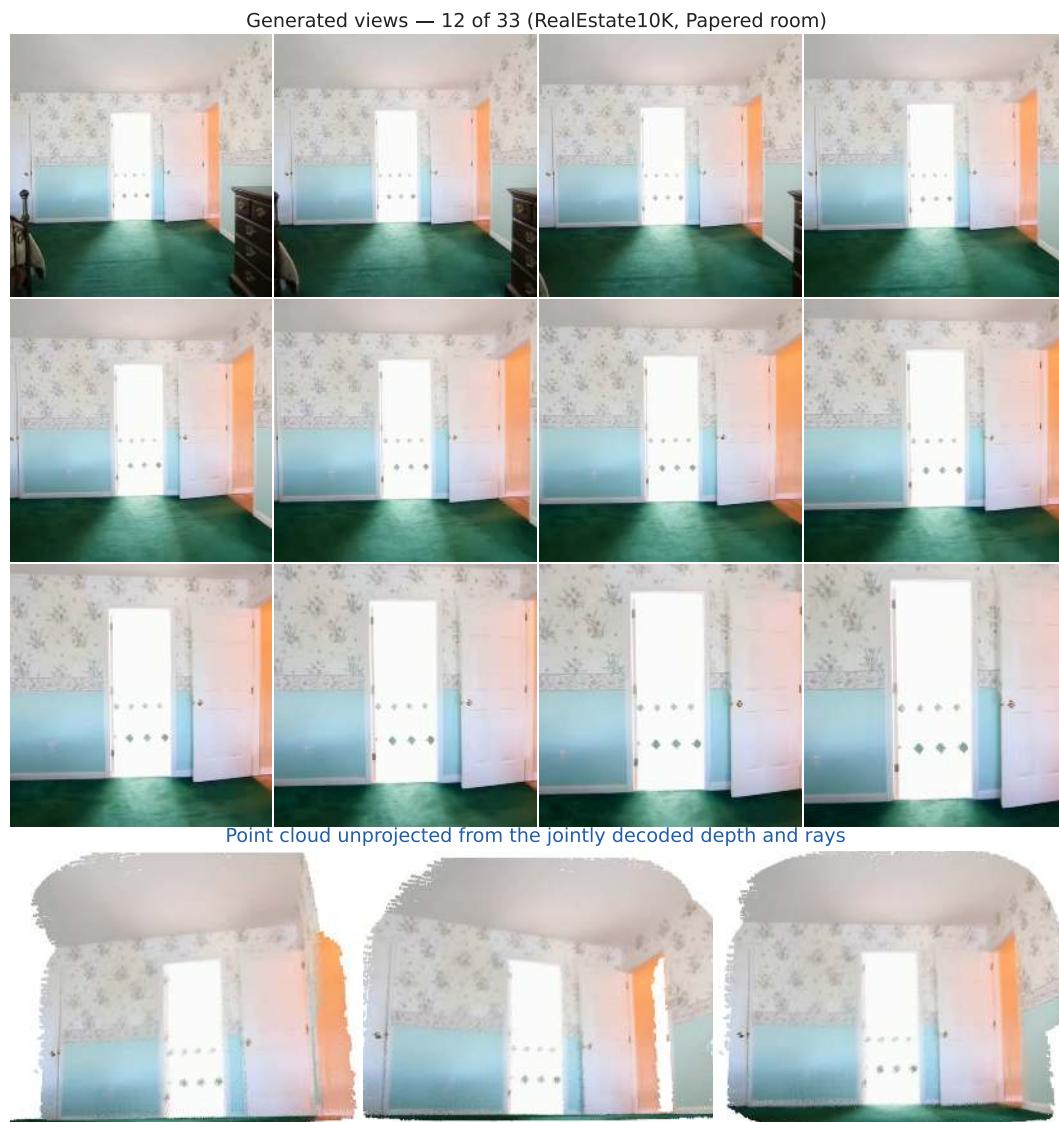


Figure 32: **Long-rollout show case: RealEstate10K papered room.** A slow pan across a papered wall to an open doorway and a green carpet. The wallpaper is a strong repeated texture on an otherwise plain surface — the case where an appearance-only latent tends to paste depth flat — yet the two side walls and the doorway recess stay on separate planes and the floor stays flat. This is the best-agreeing panel in the pool at 0.007 AbsRel.

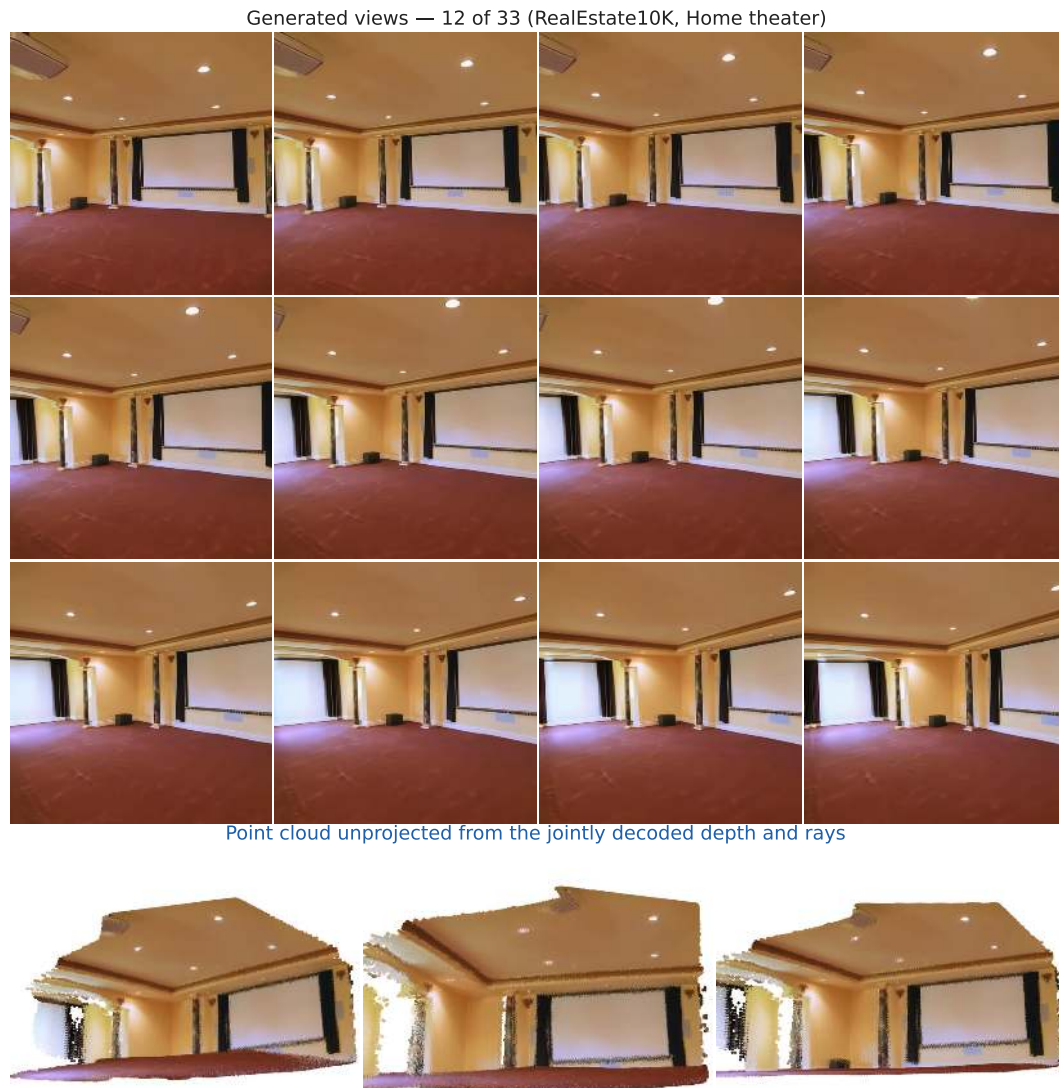


Figure 33: **Long-rollout show case: RealEstate10K home theater.** A pan across a media room with a large projection screen. The screen is a big low-texture rectangle and the drapes beside it are near-black, both weak depth cues, but the ceiling, the screen wall and the side walls close into a box in the cloud and the recessed downlights stay on the ceiling plane.



Figure 34: **Long-rollout show case: ScanNet++ office desk.** The most cluttered scene shown: the camera swings around a desk from one end to the other. Bottles, frames and the monitor stay individually resolved in the cloud instead of merging into the desk surface, the shelf under the desk keeps its two levels, and the chair separates from both. Clutter at close range is where a compact latent has the least room to be sloppy, and ScanNet++ is also the domain with the best depth agreement of the four in this pool.

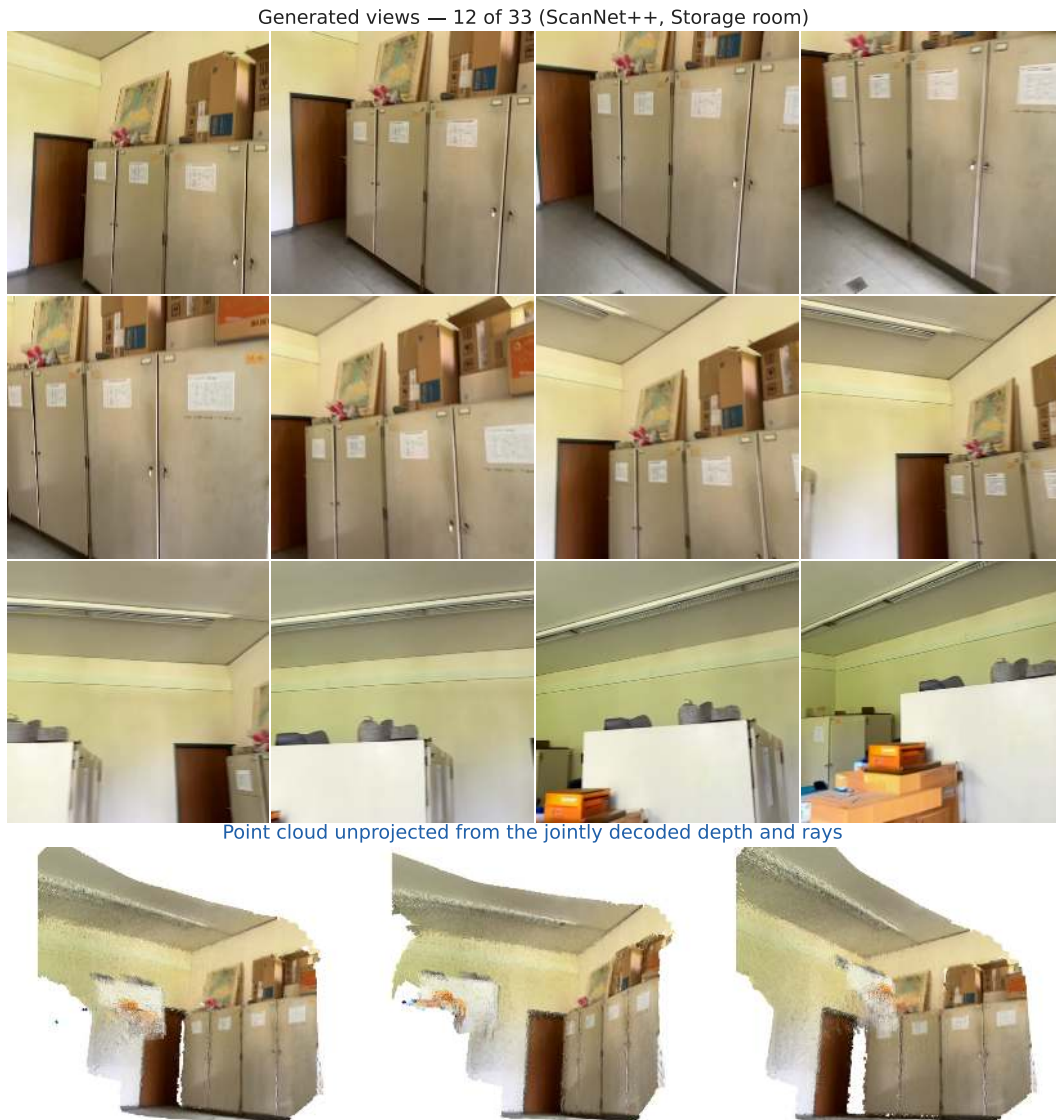


Figure 35: **Long-rollout show case: ScanNet++ storage room.** An orbit past a row of metal storage cabinets with boxes stacked on top. The cabinet faces stay on one plane with the boxes standing proud of it, the doorway to the left keeps its recess, and the ceiling-wall corner stays square — flat man-made surfaces at close range, where a small depth error would visibly bow the cabinet fronts.

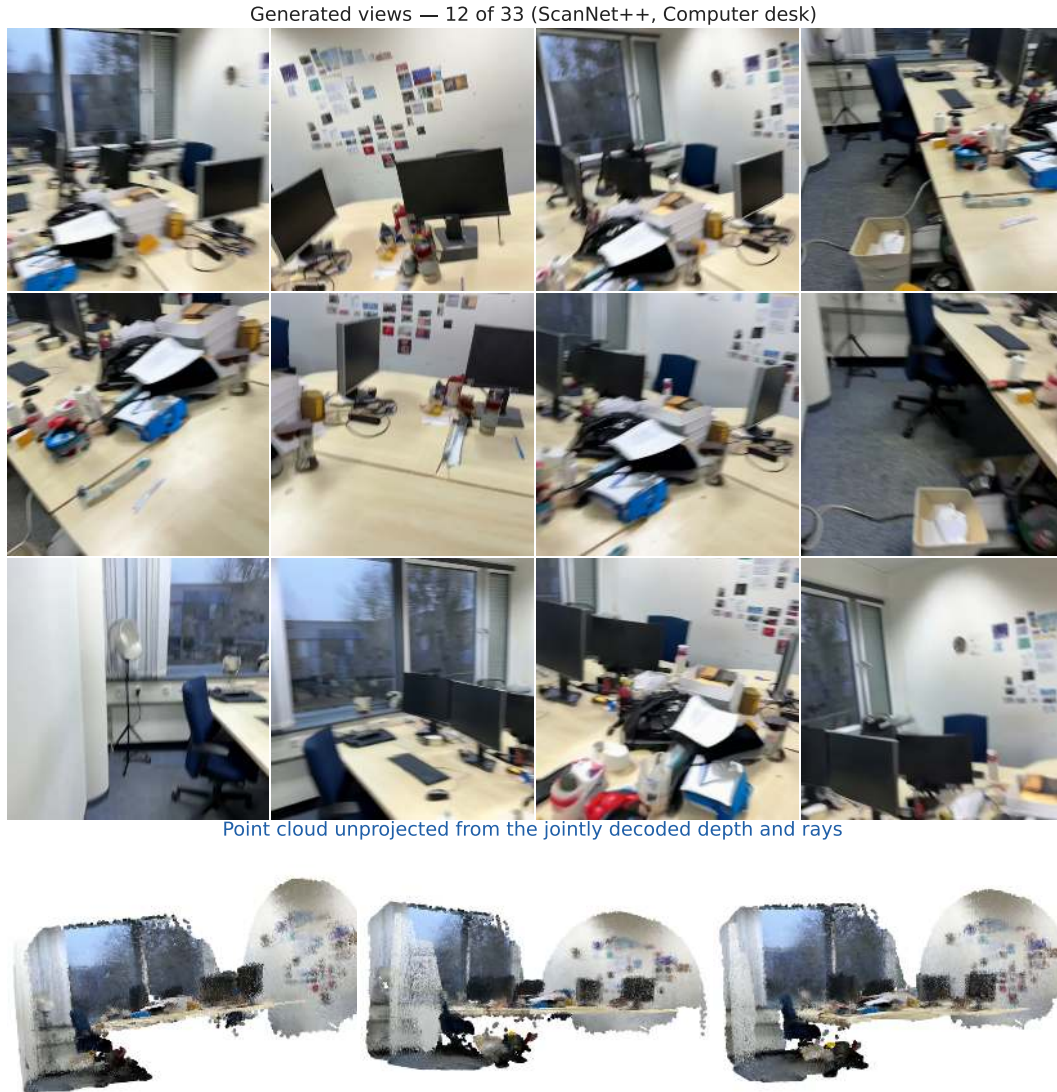


Figure 36: **Long-rollout show case: ScanNet++ computer desk.** A busy office corner: two desks with monitors, chairs, a photo-covered pinboard and a wall of windows. The monitors and chairs stay as separate volumes off the desk, the two window walls meet at a corner, and the pinboard stays flat on the wall. The glazing is the noisiest surface, as bright windows give the depth head little to lock onto.



Figure 37: Long-rollout show case: MVS-Synth city block. The one synthetic outdoor scene, and the weakest of the twelve in geometry (0.080 AbsRel, 1.7°): the generated views hold a coherent city block through a long drive down a street, but the cloud thins out where the street runs to the horizon, which is where a monocular depth decode has the least to work with. Sky-coloured points are dropped and the cloud is cropped along the street axis to the near block. The first two views sit above the block and look down at about 25° , standing the facade up with the road and crosswalk in front; the third is a near-top-down that lays the crossing, the crosswalks and the road markings out as a street plan.